Routine health information system and health facility data for neglected tropical diseases

Rabies

World Health Organization

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ISBN 978-92-4-010730-4 (electronic version)
ISBN 978-92-4-010731-1 (print version)

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Cataloguing-in-Publication (CIP) data. CIP data are available at https://iris.who.int/.

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Design and layout by L'IV Com Sàrl.

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Acknowledgements

The World Health Organization (WHO) is grateful to the following individuals who contributed to the development of this document. It was prepared in 2024.

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Abbreviations

eRIG	equine rabies immunoglobulin
hRIG	human rabies immunoglobulin
ID	intradermal
IM	intramuscular
NTD	neglected tropical disease
PEP	post-exposure prophylaxis
PrEP	pre-exposure prophylaxis
RIG	rabies immunoglobulin
RmAbs	rabies monoclonal antibodies
WHO	World Health Organization

1. Introduction

Rabies is a vaccine-preventable, zoonotic, viral disease. Once clinical symptoms appear, the disease is virtually 100% fatal. Rabies is present on all continents, except Antarctica, with over 95% of human deaths occurring in the Asia and Africa regions. In up to 99% of cases, domestic dogs are responsible for transmitting the rabies virus to humans. Rabies affects both domestic and wild animals. In the Region of the Americas, bats are now the major source of human rabies deaths as dog-mediated transmission has mostly been interrupted in this region. Bat lyssaviruses are also an emerging public health threat in Australia and western Europe. Rabies is spread to people and animals predominantly through bites or scratches, usually via saliva. Transmission can also occur if saliva of infected animals comes into direct contact with human mucosa or fresh skin wounds.

Post-exposure prophylaxis (PEP) is the emergency response to a rabies exposure. PEP includes immediate extensive wound washing (15 min with copious amounts of water and soap/detergent), a course of rabies vaccine and rabies immunoglobulin, if indicated, depending on the clinical categorization of the animal bite (I, II, III). PEP prevents virus entry into the central nervous system, which results in imminent death. Recommendations on PEP are provided in Annex 1. Although effective human vaccines and immunoglobulins exist for rabies, they are not readily available or accessible to those in need.

Rabies is a neglected tropical disease (NTD) that predominantly affects impoverished and vulnerable populations; children aged 5–14 years are frequent victims. However, despite being included in disease surveillance frameworks as a notifiable disease, globally rabies deaths are rarely reported. There are several reasons for this and some of the most common obstacles are:

- competing priorities (epidemics and pandemics, diseases with a greater economic influence);
- absence of an established surveillance system/system for collating data and lack of gathering data from local to national level;
- no designated focal point for national reporting, lack of responsibility;
- lack of data submission by human and veterinary sectors and their systems not being connected;
- incompatible, non-standardized surveillance systems and reluctance to share data between sectors as well as globally (e.g. concerns "to look bad");
- not using harmonized case definitions/no standardized data collection; and
- misdiagnosis with other diseases due to lack of knowledge or capacities to perform testing and investigations and inability to identify or investigate the biting animal.

1.1 Approach to development of this guidance

This guidance was developed based on a template provided to health programmes by the WHO Division of Data, Analytics and Delivery for Impact. WHO disease focal points in the Global Neglected Tropical Diseases Programme provided technical content in alignment with the strategic priorities presented in *Ending the neglect to attain the Sustainable Development Goals: a road map for neglected tropical diseases 2021−2030* ("the road map") *(1)* and the recommendations included in the *Rabies post-exposure prophylaxis decision tree (2)*. A drafting team comprising WHO technical officers and disease experts working on rabies at global, regional and country levels contributed to the writing and review of initial drafts of this document. Independent rabies experts reviewed the document for technical accuracy and relevance.

1.2 Declarations of interest

Declarations of interest were submitted by all contributors external to WHO for review. No significant interests were identified.

1.3 Scope of this document

This document mainly addresses the collection and analysis of individual data on the treatment of rabies exposures, including PEP, and rabies cases collected at rabies treatment centres/health facilities. It also offers guidance on using data to assess the risk of exposure when local capacity to assess animals for rabies, and laboratory capacity to test and report results within several days, are available. In certain circumstances, when the risk of rabies is very low, PEP can be delayed pending the outcome of an animal assessment, confinement period or testing, or even be discontinued.

Data not specifically addressed in this document but nonetheless important for successful rabies elimination programmes using a One Health approach are those related to:

- animal follow-up as a result of an integrated bite case management programme;
- rabies cases in animals, dog population and dog vaccination coverage; and
- rabies pre-exposure prophylaxis (PrEP).

1.4 Objectives

This document provides guidance on the analysis and use of routine rabies data collected at the health facility level. It presents core facility indicators and analysis, provides suggestions for questions on data quality as well as considerations and limitations for using the data and analysis. By the end of this document, readers should be able to:

- describe core rabies indicators and notable trends in rabies prevalence, geographical distribution of the disease and progress towards disease control;
- monitor the programme performance of rabies diagnosis and animal exposure case management;
- understand how to interpret changes in trends over time or by geography; and
- assess data quality and understand its implications when interpreting data.

1.5 Target audience

This document is relevant for different members of the health workforce working on rabies including:

- health workers at health facilities treating animal bite victims and diagnosing rabies cases in clinical settings;
- ministry of health decision-makers, epidemiologists and data managers working in the national rabies programmes, and health information system managers at district and higher levels;
- staff of partner organizations supporting the strengthening of national rabies programmes, the management of animal bite cases and the diagnosis of rabies cases in certain health facilities or health system strengthening; and
- consultants and staff working at research institutes involved with the assessment and improvement of rabies control activities and specifically with the analysis of rabies data.

2. About rabies data

2.1 Importance of data on human rabies disease

The road map sets global targets for 2030 and milestones to prevent, control, eliminate and eradicate a diverse set of diseases, including rabies *(1)*. In line with the road map, WHO together with the World Organisation for Animal Health, the Food and Agriculture Organization of the United Nations and the Global Alliance for Rabies Control published a global strategy for the goal of zero human dog-mediated rabies deaths by 2030, worldwide, or "zero by 30" *(3)*.

Human deaths from rabies are significantly underreported in many parts of the world. Based on current estimates *(4)*, about 44 000 human deaths annually in over 150 countries are due to rabies, of which less than 5% of cases are reported to international health authorities. Systems for the detection, investigation and reporting of human and animal rabies cases are lacking in many countries. Rabies control efforts remain fragmented and uncoordinated in many areas, resulting in unclear responsibilities and assumptions about other stakeholders doing something, although in reality neither does or duplicates efforts, inefficient use of resources and ineffective or unsustainable activities. Without accurate data, the real burden of rabies on communities is unknown, which impedes obtaining political and financial commitment as well as country ownership. Improved data collection and evaluation also provides evidence for informing rabies management decisions, including PEP policies, mass dog vaccination approaches and community educational programmes. Understanding disease epidemiology and dog population dynamics are crucial for targeted action as well as for judicious use of PEP. Evidence-based health programmes start with appropriate data collection systems, which can inform evaluation of key indicators, thereby enabling stakeholders to adapt and adjust their approach to rabies elimination and develop stronger elimination programmes. Monitoring of rabies programmes and disease surveillance are needed to measure impact and progress towards the global goal of Zero by 30 and for increasing awareness and advocacy.

This document covers the use of data that can be routinely collected and reported by health facilities to aid in establishing evidence-based rabies control, prevention and surveillance systems. Standardized case definitions and rabies indicators allow for reliable and comparable data on all levels. This helps to strengthen not only the health system but also to foster collaboration using a One Health approach. The template for case-based data collection that is included in this guide can be used to conduct risk assessments at the health facility level, provide guidance for PEP treatment for individual cases and keep track of patient visits to monitor PEP compliance.

2.2 Objectives of surveillance data

Collection of good-quality data allows programme managers to analyse, adjust and evaluate ongoing rabies programmes. Surveillance data can be used (this is not a finite list) to:

- Identify high prevalence geographical locations and at-risk communities so as to:
  - assign priorities, focus responses, allow rapid interventions
  - direct resource allocation to population in need
  - assess geographical location of anti-rabies clinics
  - identify areas to conduct mass dog vaccinations
  - raise awareness in at-risk-communities, increase community support

  - trigger follow-up investigations (integrated bite case management with timely sharing of data)
  - estimate disease burden
  - identify endemicity or validate rabies-free areas by implementing surveillance throughout the entire country
- Provide information for guiding interventions so as to:
  - track patient treatment outcomes including PEP completion and adverse events
  - track progress of public health programme control efforts
  - allow early warning system/alerts to recognize outbreaks/increased prevalence
- Forecast needs (e.g. for biologicals, diagnostics, logistic requirements), especially to:
  - forecast for vaccine procurement, storage, and budgeting
  - forecast for immunoglobulin production, storage and budgeting
  - assure supplies of vaccine and immunoglobulin
- Assess progress through monitoring and evaluation, meaning to:
  - assess and validate the health programme at all levels: subnational, national, regional and global
  - locate important geographical areas, population groups, trends over time
  - analyse how and why targets are being achieved or not
  - adjust programmes and increase performance and efforts
  - ensure evidence and data driven policy and planning
  - use as advocacy for funding and resource mobilization
  - support the global rabies elimination plan
  - validate rabies-free zones
- Review and evaluate health system performance in terms of:
  - service delivery/accessibility
  - compliance
  - effectiveness
  - efficiency
  - sustainability
  - quantification of contributions to health system strengthening and intersectoral collaboration
  - engaging leadership and governance performance
- Understand drivers of disease more fully (including co-morbidities)

2.3 WHO-recommended rabies-related definitions

2.3.1 Human rabies case

A human rabies case is a person with an acute neurological syndrome (i.e. encephalitis) dominated by forms of hyperactivity (furious rabies) or paralytic syndromes (paralytic rabies) progressing towards coma and death, usually by cardiac or respiratory failure, typically within 7–10 days of the first signs if no intensive care is instituted. These may include:

- aerophobia,
- hydrophobia,
- paraesthesia or localized pain,
- dysphagia,
- localized weakness and
- nausea or vomiting.

2.3.2 Suspected rabies case

A suspected rabies case is a case that is compatible with a clinical case definition of human rabies.

2.3.3 Probable rabies case

A probable rabies case is a suspected case that has a reliable history of contact with a suspected,[1] probable[2] or confirmed[3] rabid animal.

2.3.4 Confirmed rabies case

A confirmed rabies case is a suspected or probable case that is confirmed in a laboratory.
Note: In the "Core indicators" section below, "rabies cases" refers to "suspected rabies cases".

2.4 Additional WHO-recommended definitions with implications for clinical management of cases

2.4.1 Categories of contact with suspected rabid animals

- Category I: touching or feeding animals, animal licks on intact skin (no exposure)
- Category II: nibbling of uncovered skin, minor scratches or abrasions without bleeding (exposure)
- Category III: single or multiple transdermal bites or scratches, contamination of mucous membrane or broken skin with saliva from animal licks, exposure due to direct contact with bats (severe exposure)

2.4.2 Number of people receiving post-exposure prophylaxis

For data collection, PEP is defined as wound care and at least 1 dose of rabies vaccine.

[1] A suspected rabid animal is defined as an animal that presents with any of the following signs: hypersalivation, paralysis, lethargy, unprovoked abnormal aggression (e.g. biting two or more people or animals and/or inanimate objects), abnormal vocalization, diurnal activity of nocturnal species.

[2] A probable rabid animal is defined as a suspected case with a reliable history of contact with a suspected, probably or confirmed rabid animal and/or an animal with suspected rabies that is killed, died or disappeared within 4–5 days of illness being observed.

[3] A confirmed rabid animal is defined as a suspected or probable case that is confirmed in a laboratory.

3. Data quality

One of the challenges of interpreting health facility data is the responsibility for data recording, entry, cleaning and management. Unlike special studies or surveys, resources for entering and cleaning data are often limited in health facilities, impacting the quality and usability of routine monitoring data for critical data categories such as human rabies cases, rabies exposures and PEP utilization. Systems and protocols must therefore be established to enhance good human rabies data collection, and reporting facilitates data analysis and use.

However, as for all data sources, any analysis must consider whether the results are affected by data quality issues. A key example can be seen when using health facility data to determine areas of high rabies risk; lack of full institutional participation can lead to areas of apparent low risk, when the reality is that data are just not collected and/or shared. Rabies data often pose several specific challenges such as inconsistent case definitions, loss of follow-up on patients, various reporting methods by the human health sector and veterinary services that may not overlap and limited regular data reporting on a national level.

The WHO data quality review toolkit *(5)* and preventive chemotherapy field manual *(6)* provide guidance for defining measures of data quality, conducting a desk review to assess data quality and conducting data verification of routine facility data systems. The principles in the toolkit govern and are applied to the data architecture on national routine health information systems in general. The five key domains used for periodic assessments of data quality are summarized below.

3.1 Completeness of data

The completeness of the data is assessed by measuring whether all the health facility level and administrative level entities which are supposed to report actually do so. This applies to health-facility reporting to districts and to district reporting to the regional or provincial/state levels.

3.2 Timeliness of data

The timeliness of data is assessed by measuring whether the health facility level and administrative level entities which submitted reports did so before a predefined deadline. This can for example be accessed via notification rates by administrative area.

3.3 Consistency of reported data

3.3.1 Internal consistency

The internal consistency of reported data relates to the coherence of the data being evaluated. Internal consistency metrics examine: (i) coherence between the same data items at different points in time; (ii) coherence between related data items at sequential points in the data flow pathway; and (iii) accuracy in comparison of data in source documents and in national databases.

3.3.2 External consistency

The external consistency of reported data refers to the agreement with other sources of rabies data such as surveys and estimates. The level of agreement between two sources of data measuring the same health indicator is assessed. Both sources of data usually compared are data flowing through the health

management information system or the programme-specific information system and data from a periodic population-based survey. The health management information system's rabies data can also be compared to the national logistics management system data, pharmacy records or other types of data to ensure that the two sources fall within a similar range.

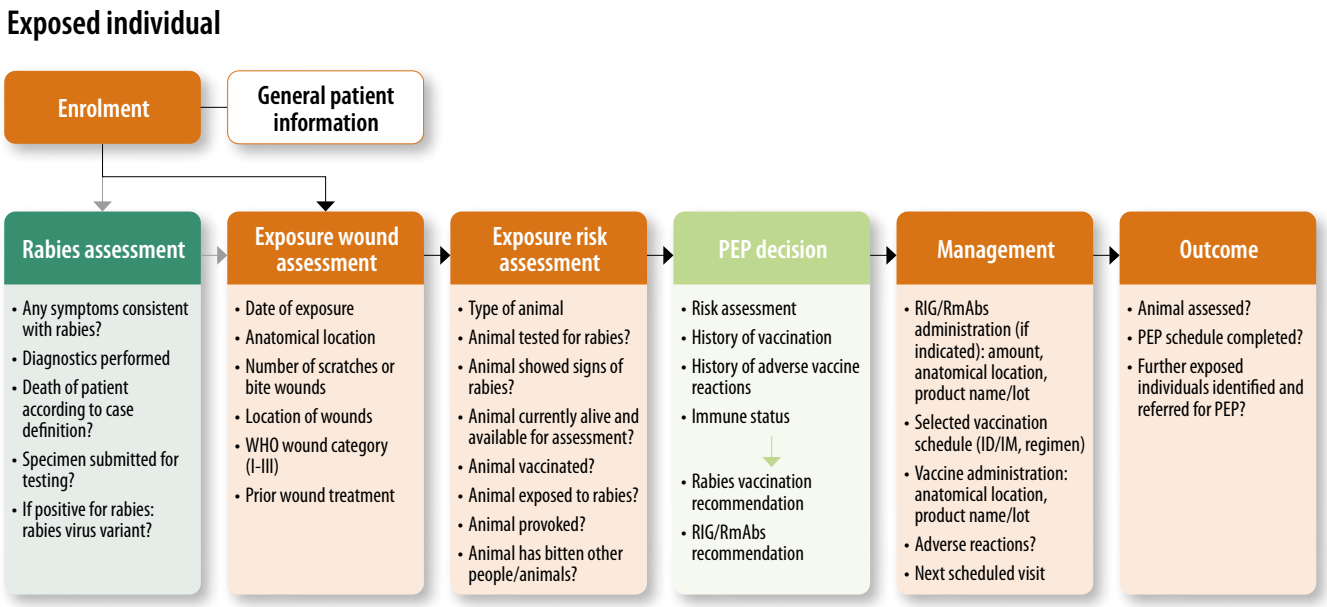
3.4 External comparisons of population data

External comparisons of population data refer to reviews of the denominator data used to calculate rates for performance indicators. Population data serve as the denominator in the calculation of a rate or proportion and provide important information on coverage. This data quality measurement compares two different sources of population estimates (for which the values are calculated differently) in order to ascertain their level of congruence. If both population estimates are discrepant, the coverage estimates for a given indicator can be very different even though the programmatic result (i.e. the number of events) is the same. The higher the level of consistency between denominators from different sources, the more likely it is that the values represent the true population value. For NTD programmes, the primary source of population data that is used in denominators is from the national census records.

4. Case-based reporting and risk assessment

Rabies mainly kills those who cannot access timely and effective PEP. Delay in seeking PEP, inadequate wound care, unrecognized wounds, exposures to the head and neck, and noncompliance of patients with vaccination schedules, among other factors, can also contribute to PEP failure and subsequent death. Conversely, irrational fear of rabies after an animal exposure may lead to detrimental costs for the families of bite victims and stock-outs of PEP. Therefore, a careful balance must be achieved to ensure that people with eligible exposures seek and receive the adequate care, without causing undue financial and health-care burdens by overusing rabies biologicals to treat exposures that do not meet the criteria for suspected rabies exposures (Fig. 1). The decision trees (Annex 2 and 3) act not only as a decision aid to assist health-care providers in determining if PEP is indicated but also collect information for surveillance reporting. This assessment can be broken down into four categories: (i) patient information, (ii) exposure wound assessment, (iii) exposure risk assessment and (iv) clinic visits.

Fig. 1. Collection and use of individual case-based data for risk assessment



ID: intradermal; IM: intramuscular; PEP: post-exposure prophylaxis; RIG: rabies immunoglobulin; RmAbs; rabies monoclonal antibodies.

4.1 Patient information

This section documents commonly collected patient information. The key fields for this tool to work correctly are the date of the first health-care visit, at which time the wound management and a risk assessment is conducted, the current health status of the patient, the patient's weight (in kilograms), history of rabies vaccination and immune status. These fields are essential for reporting and appropriate PEP administration when indicated. Patient contact information should also be collected, to ensure that any animal assessment or testing outcomes are communicated back to the patient.

The health history section assesses patient weight for rabies immunoglobulin (RIG) calculations, previous vaccination history and immune status. This information aids in the determination of the specific vaccination schedule for the patient. Sometimes, a patient may present to the health facility and human rabies is suspected. Patients suspected of having active rabies disease should not receive PEP, as this provides no improvement in clinical outcome and can hasten the time to death.

4.2 Exposure wound risk assessment

To assess the type of wound experienced by the patient, questions pertaining to the date of exposure, anatomical location and number of wounds as well as the wound categorization of the exposure are entered into the form. Based on the responses entered, PEP recommendations will be made. The respective decision tree can be found in Annex 2.

4.3 Post-exposure prophylaxis risk assessment

The PEP risk assessment uses factors about the offending animal such as species, health status, number of other animals and people affected, and availability for veterinary assessment, quarantine or testing to determine if the event is consistent with the WHO definition of a suspected rabies exposure. Many risk assessment factors may not be known at the time of risk assessment; the logic framework treats unknown responses as equivalent to the highest risk option. In deciding the health status of the offending animal, particularly when this is an unknown or stray animal, health-care providers should take a more conservative approach by defaulting to unknown unless there is ample evidence to support the animal being deemed healthy. Additionally, when considering if the animal was exposed to rabies in the past 12 months, if the animal is allowed to roam freely unsupervised, health-care providers should default to unknown. If the vaccination status of the animal is uncertain, “no or unknown” should be selected.

At the conclusion of this section, the health-care provider will be provided with a Risk Assessment Determination. Additional recommendations for RIG/RmAbs and antiretrovirals are included based on information from the wound risk assessment. PEP can either be not indicated, should be started immediately or can be delayed pending the outcome of animal observation or testing. The respective decision tree can be found in Annex 3.

Note: Health-care providers should consider the PEP recommendations derived from the decision tree as a recommendation; however, the final decision on if PEP is indicated should be determined by the health-care provider, in consultation with the patient, public health department and veterinary services (when applicable).

4.4 Clinic visit

The clinic visit section allows the health-care provider to ensure that the patient is on schedule as well as to track the biological administered, type of products used and document any adverse reactions. The decision tree will also ensure appropriate administration of rabies biologicals. It accomplishes this by calculating the amount of RIG/RmAbs the patient should receive based on the patient weight entered in the patient information section, informing about correct anatomical locations for vaccine administration as well as notifying the health-care provider if vaccine and RIG/RmAbs is administered in the same anatomical site, which should never occur and would invalidate both applications. After the schedule is selected, the date of the next visit will populate. In the event of a deviation, the algorithm will recalculate the dates to get the patient back on the correct dosing interval.

Patients requiring repeat visits should be enrolled in a tracking system to ensure continuity of care. Patient vaccination tracking systems can be variable, from using scheduling cards to electronic systems, and active patient follow-up is suggested.

A detailed, hands-on step-by-step protocol on providing WHO-recommended rabies PEP for patients including decision-making process and frequently asked questions has been published *(7)*, including a printer-friendly supplementary version *(8)*.

5. Indicators

5.1 Core indicators

The recommended minimum core indicators, definitions, disaggregation and data sources for rabies surveillance are provided in Table 1. Obtaining data from individual rabies case registers is preferred over aggregate forms to ensure high data quality.

Table 1. Rabies core indicators

Core indicators	Definition	Disaggregation	Comments
Total number of human rabies cases (ICD-11 codes: 1C82, 1C80)	Total number of reported human rabies cases during the reporting period (i.e. in one year)	• Sex (male/female) • Age* • Transmission (dog, bat, other) • Confirmation status (clinical or laboratory diagnosis)	Key indicator for rabies burden, DALYS, case rate, testing rate, trend analysis to monitor progress Animal-mediated rabies cases
Total number of dog-mediated human rabies cases	Total number of reported dog-mediated human rabies cases during the reporting period (i.e. in one year)		Global target to end human deaths from dog-mediated rabies by 2030
Proportion of dog-mediated human rabies cases	Proportion of dog-mediated human rabies deaths during the reporting period (i.e. in one year)		Percentage of dog-mediated cases
Human rabies mortality rate	Total number of human rabies deaths per 10 000 population during the reporting period (i.e. in one year)	• Sex (male/female) • Age* • Transmission (dog, cat, bat, wildlife, livestock)	
Dog-mediated human rabies mortality rate	Total number of dog-transmitted human rabies deaths per 10 000 population during the reporting period (i.e. in one year)		
Proportion of laboratory-confirmed rabies cases	Proportion of laboratory-confirmed cases during the reporting period (I.e. in one year)		Diagnostic capacity
Number of reported animal bite cases in humans[1]	Total number of people exposed to rabies susceptible animals during the reporting period (i.e. in one year)	• Sex (male/female), • Age* • Species of exposed animal (dog, cat, bat, wildlife, livestock) • WHO wound category (I, II, III)	Key indicator for exposure rate, vaccine forecasting, trend analysis, source of exposure, track burden, inform bite prevention campaigns, trigger follow-up investigations
Number of people receiving PEP (ICD-11 code: QC01.2)	Total number of people receiving at least one PEP vaccination dose during the reporting period (i.e. in one year)	• Sex (male/female) • Age* • Exposure category (I, II, III)	

[1] Disaggregation by WHO wound category also includes of mucous membranes exposures or scratches.

Table 1. *contd.*

Core indicators	Definition	Disaggregation	Comments
Proportion of animal scratch/bite victims receiving PEP	Proportion of exposed victims receiving at least one PEP vaccination dose during the reporting period (i.e. in one year)	• Sex (male/female) • Age* • Exposure category (I, II, III)	Key indicator for vaccine forecasting, awareness, trend analysis, efficiency of PEP utilization
Proportion human rabies cases following an(y) animal exposure	Proportion of people developing rabies after being scratched or bitten by animals during the reporting period (i.e. in one year)	• Sex (male/female) • Age* • Exposure category (I, II, III) • Species of offending animal (dog, cat, bat, wildlife, livestock)	
Proportion of rabies cases following a dog exposure	Proportion of people developing rabies after being scratched or bitten by a dog during the reporting period (i.e. in one year)		

* Recommended age groups for global annual NTD reporting: By age group (years): <1 year, 1–4 years, 5–14 years, 15–24 years, 25–49 years, 50–64 years, 65 years and older. Minimum age groups for other reporting: 0–4 years, 5–14 years, 15 years and older.

DALYs: disability-adjusted life-years; ICD-11: International Classification of Diseases, 11th edition; PEP: post-exposure prophylaxis; RIG: rabies immunoglobulin; RmAbs; rabies monoclonal antibodies.

5.2 Additional indicators

Additional rabies elimination programme indicators that can be generated/calculated based on the case-based recording form are provided in Table 2. A glossary for case-based reporting form is provided in Annex 4.

Table 2. Additional rabies indicators

Additional indicators	Definition	Disaggregation	Comments
Geographical distribution where animal exposures occurred	Geographical distribution of where exposures are occurring	• Village/ district/region • Species of offending animal (dog, cat, bat, wildlife, livestock) • Exposure date	For tracking, identify risk areas / hot spots, seasons and at-risk communities, target interventions and campaigns
Healthcare facility catchment area	How far did the patient travel to reach the clinic (hours of travel)?	• Village/district/region	Access to PEP, service coverage
Total number of exposed victims receiving RIG or RmAbs	Total number of exposed victims that received RIG/RmAbs	• Sex (male/female) • Age* • WHO wound category (I, II, III) • Species of offending animal (dog, cat, bat, wildlife, livestock) • Type of rabies biological (RIG, RmAbs)	Availability
Proportion of category III bite victims that receive RIG/RmAbs	Proportion of people with WHO wound category III exposure receiving RIG	• Sex (male/female) • Age* • Species of offending animal (dog, cat, bat, wildlife, livestock)	

Table 2. *contd.*

Additional indicators	Definition	Disaggregation	Comments
Proportion of exposed victims receiving PEP completing their vaccination schedule	Proportion of exposed victims that completed their vaccination schedule	• Sex (male/female) • Age* • WHO wound category (I, II, III) • Species of offending animal (dog, cat, bat, wildlife, livestock) • Vaccination scheme (IM/ID) • Anatomical location of the scratch/bite	Compliance, service coverage, outreach, loss of follow up
Proportion of people showing adverse events to vaccination	Proportion of people that showed adverse events after receiving vaccination/RIG	• Sex (male/female) • Age* • Reaction type? • Suspected cause of reaction (RIG, RmAbs or vaccine)	PEP schedule recommendation, tracking of adverse events
Rabies incubation period	Time between animal exposure and (clinical) rabies diagnosis	• Species of biting animal (dog, cat, bat, wildlife, livestock) • Exposure category (I, II, III) • Location of the bite • PEP status	The incubation period for rabies is typically 2–3 months but may vary from 1 week to 1 year, dependent upon factors such as the location of virus entry and viral load. Enables clinicians to consider rabies as a differential in neurological and psychiatric diagnoses.
Rabies virus variant	Number of laboratory-confirmed rabies cases by the rabies virus variant	• Dog/bat/terrestrial carnivore (wildlife)/other	Needed to confirm canine-rabies freedom
PEP vaccination scheme	Overview of the used vaccination sites and regimens	• IM/ID • Anatomical location of vaccine administration	Service coverage, vaccine usage, forecasting
Anatomical location of the scratch/bite	Overview of the anatomical location of the scratch/bite	• Species of offending animal • Exposure category (I, II, III)	Identify high-risk exposures (multiple bite wounds, deep wounds, bites to head and highly innervated parts of the body)

* Recommended age groups for global annual NTD reporting: By age group (years): <1 year, 1–4 years, 5–14 years, 15–24 years, 25–49 years, 50–64 years, 65 years and older. Minimum age groups for other reporting: 0–4 years, 5–14 years, 15 years and older.

ID: intradermal; IM: intramuscular; PEP: post-exposure prophylaxis; RIG: rabies immunoglobulin; RmAbs; rabies monoclonal antibodies.

Secondary data to be collected from veterinary health services on animal investigations, animal rabies cases, dog population and dog vaccination rates by administrative area are described in Annex 5.

6. Core analyses

When analysing data over time, check whether the values increase or decrease, and how the change compares with what is expected based on initial (baseline) values, the targeted values, and the quality and location of interventions conducted. Assess whether the changes are uniform across locations; if not, check whether there are issues with access or availability of services in particular locations. This information should be used to inform how interventions are carried out. Disease surveillance systems for rabies should reflect information sharing with veterinary health systems, especially for case investigations of biting animals.

For rabies programmes that are in the early stages of implementation, changes in data may be reflective of new data collection activities and are not necessarily an indication of a worsening rabies situation. When analysing rabies surveillance data, a lack of reports may not reflect low-risk; any area with a lower prevalence of rabies (human or animal) should first be assessed for data quality. Reporting data also includes reporting "0" numbers to allow a distinction between no data and real o values. If data quality is suitable and eligible institutions are reporting rabies case data, then these areas may truly reflect low-risk.

Finally, rabies case data are difficult to interpret without an appropriate denominator; for many analyses, this requires reporting of negative test data and the number of patients assessed for a potential rabies exposure (regardless of whether PEP was recommended or not).

6.1 Epidemiology of rabies disease

This section provides a brief description of each of the core analyses that are recommended on the epidemiology of rabies, and factors to take into consideration in their interpretation. Before interpreting the trends in the epidemiological data, it is key to investigate issues of data quality. If you see any sudden changes in trends, first check data to make sure there has not been any error in data entry.

6.1.1 Total number of human rabies cases

Rabies should be a notifiable disease and each patient falling under WHO's rabies case definitions has to be reported and investigated.

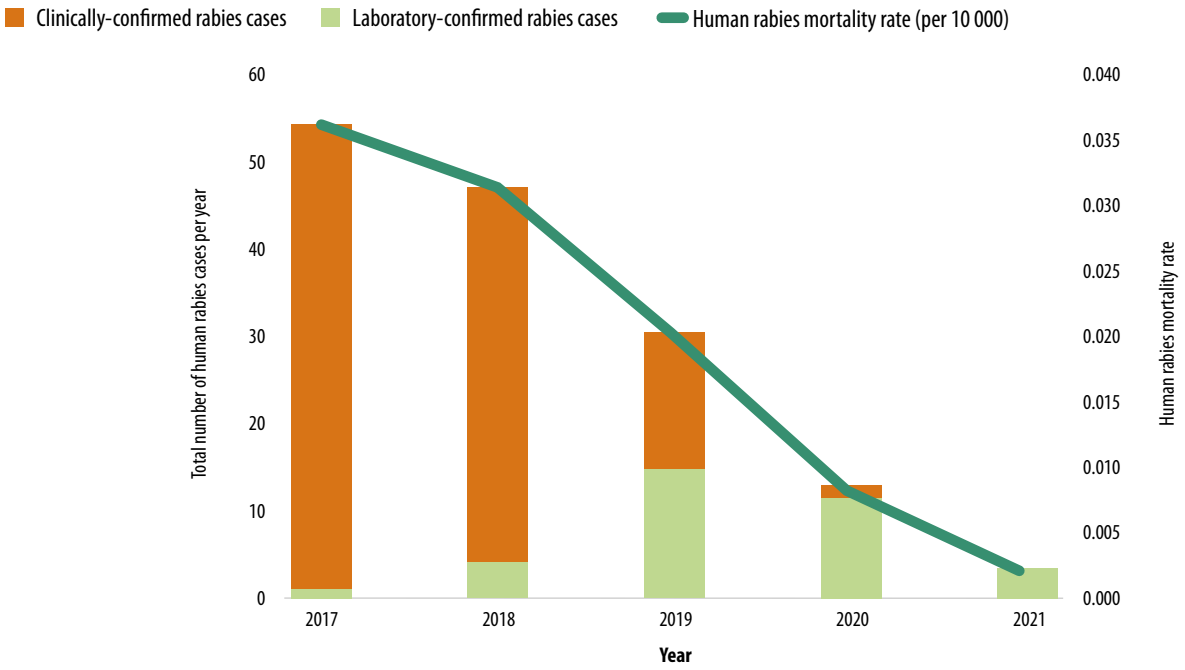
Rabies case numbers (either clinically or laboratory diagnosed) are the main indicator that helps to assess the burden of rabies and monitor the impact of interventions and public health programme efforts. Ideally, cases can be disaggregated by sex, age, as well as by the animal that transmitted the disease, informing decision-making processes and intervention design.

6.1.2 Human rabies mortality rate

Number of human rabies deaths / population of administrative area × 10 000

Informs about the burden by measuring the frequency of death by rabies scaled to the size of the population in a defined area. It shows how many deaths in the specified population are attributed to rabies (Fig. 2).

Fig. 2. Total number of human rabies cases per year (left y-axis) disaggregated by confirmation status in line with the WHO case definition (see section 2.3 for *WHO-recommended rabies-related definitions*) and human rabies mortality rate (right y-axis). Rabies cases decreased over the past 5 years and proportionally more and more cases were confirmed laboratory.



Source: WHO Global Health Observatory *(4)*.

6.1.3 Proportion of human rabies cases following an(y) animal bite/animal exposure

Total number of human rabies cases / Total number of humans bitten by an animal × 100

Gives an overview of how many people that have been wounded by an animal develop rabies. Disaggregation should ideally include sex, age, species of the animal that wounded the patient and WHO wound category.

Focusing on dog-transmitted rabies

Considering the global goal of zero dog-mediated human rabies by 2030 it is important to know how many cases have been transmitted by dogs. Hence, disease indicators should be disaggregated by transmission source.

6.1.4 Total number of dog-mediated human rabies cases

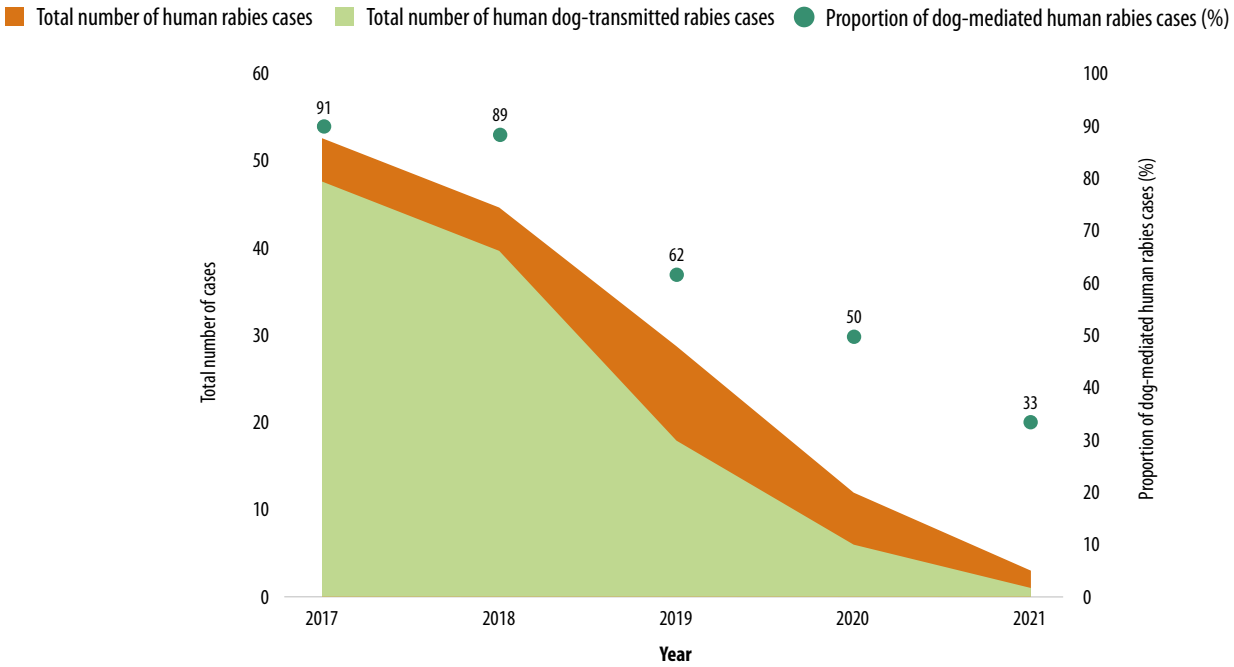
Disaggregation of case numbers by animal source is crucial for monitoring progress to end human deaths from dog-mediated rabies. Tracking cases and case investigations are especially important in the elimination phase. To validate elimination of rabies as a public health problem, countries need to demonstrate that there has been no human dog-mediated rabies death for at least 24 months.

6.1.5 Proportion of dog-mediated human rabies cases

Number of dog-mediated human rabies cases / Total number of human rabies cases × 100

Indicates the percentage of rabies cases that have been transmitted by a dog (Fig. 3).

Fig. 3. **Proportion of dog-mediated human rabies cases in comparison with all human rabies cases. The number of dog-transmitted human-rabies cases has decreased over the past years, suggesting successful implementation of programmes and measures.**



Source: WHO Global Health Observatory *(4)*.

6.1.6 Dog-mediated human rabies mortality rate

Number of dog-transmitted human rabies deaths / population of administrative area × 10 000

Specifies the burden of dog-transmitted rabies in a defined population by showing how many deaths in the specified population are attributed to dog-transmitted rabies.

6.1.7 Proportion of rabies cases following a dog bite/an exposure to a dog

Number of dog-mediated human rabies cases / Total number of humans bitten by dogs × 100

Gives an overview of how many people that have been wounded by a dog develop rabies.

6.1.8 Additional analysis

Geographical distribution where animal bites occurred (rural vs urban, hot spots)

Animal exposures (scratches, bites) can be mapped in accordance with their geographical occurrence (e.g. village, district, region), disaggregated by the species of the animal and the exposure date.

This mapping allows identification of where and which kind of animal poses a risk to humans and assists in targeting responses and allocating resources directly to the population in need (e.g. mass dog vaccination and education campaigns). It also helps to identify areas where anti-rabies clinics should be established and to distribute rabies biologicals.

Geographical information can further be used to validate rabies-free areas on the way to eliminating rabies.

Rabies incubation period

Time between exposure and onset of disease (usually calculated after clinical diagnosis of the disease). This information should be accompanied by the species of the biting animal, the WHO exposure category, the location of the bite (anatomical location, single or multiple wounds) and the rabies vaccination status of the victim (PrEP or PEP received, how many doses, wound care, RIG application).

6.2 Clinical examination

This section provides a brief description of key clinical findings to observe and record.

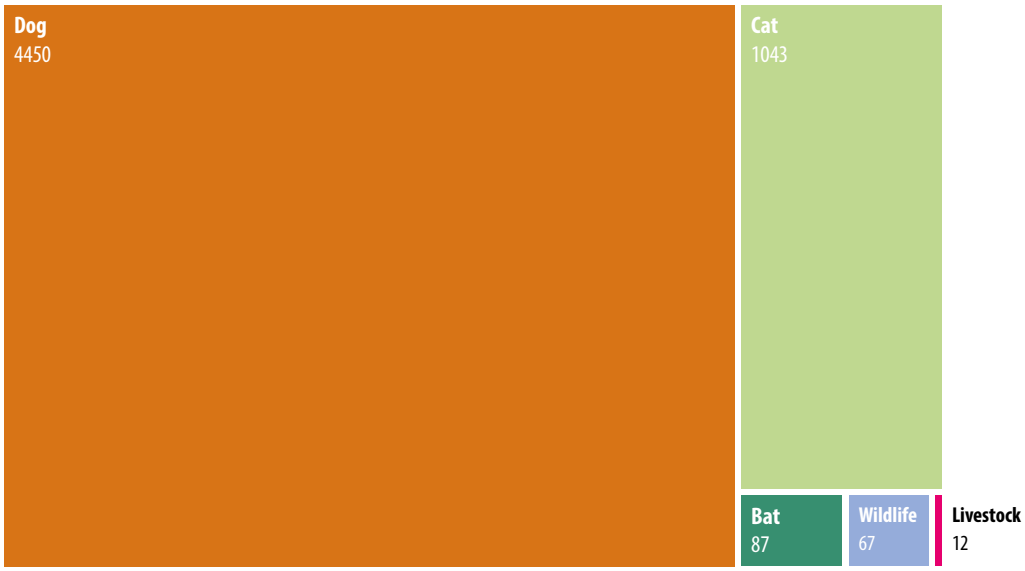
6.2.1 Number of reported animal bite cases/animal exposures in humans

Ideally disaggregated by sex, age, species of biting animal and WHO wound category (Fig. 4 and 5). The number is a key indicator for exposure rate. It provides information on the burden of exposures including the source of animals susceptible to rabies and helps to inform countermeasures (e.g. bite prevention campaigns) and track their success (analysing the trend over time). It further informs forecasting for procurement, storage, and budgeting of rabies biologicals to ensure adequate supplies of vaccine and immunoglobulin.

Fig. 4. Total number of animal exposures in a year disaggregated by WHO wound category. The graph offers information on the amount of exposures including severity (Category II vs Category III).

Source: WHO Global Health Observatory *(4)*.

Fig. 5. Disaggregation of number of animal exposures by animal in a year. The visualization shows that most of the exposures come from dogs then cats followed by bats, wildlife (racoons) and livestock.



Source: WHO Global Health Observatory *(4)*.

6.2.2 Additional analysis

Location of the bite/wound as well as distribution of single or multiple bites/wounds

This advanced information can be used to assess the risk of a bite victim and to prioritize for example RIG in case of shortage. Deep, multiple bite wounds as well as bites to head and other highly innervated parts of the body are considered high-risk bites.

Proportion of bite victims/exposed individuals classified as at-risk

If advanced risk assessment of bite victims has been established, information on how many bite victims were finally classified as at-risk can be used to inform decisions and policy making as well as forecasting of rabies biologicals. This information can also be used to develop education materials to better inform at-risk populations on the risks associated with animal bites.

Proportion of bite victims/exposed individuals where animal could be assessed

Information about the assessment of biting animals is important to establish integrated bite case management programmes and helps to inform decision making. To monitor efforts, it is important to know how many animals have been assessed and to adjust communication channels and capacity if the number does not reach a satisfying target.

6.3 Diagnosis

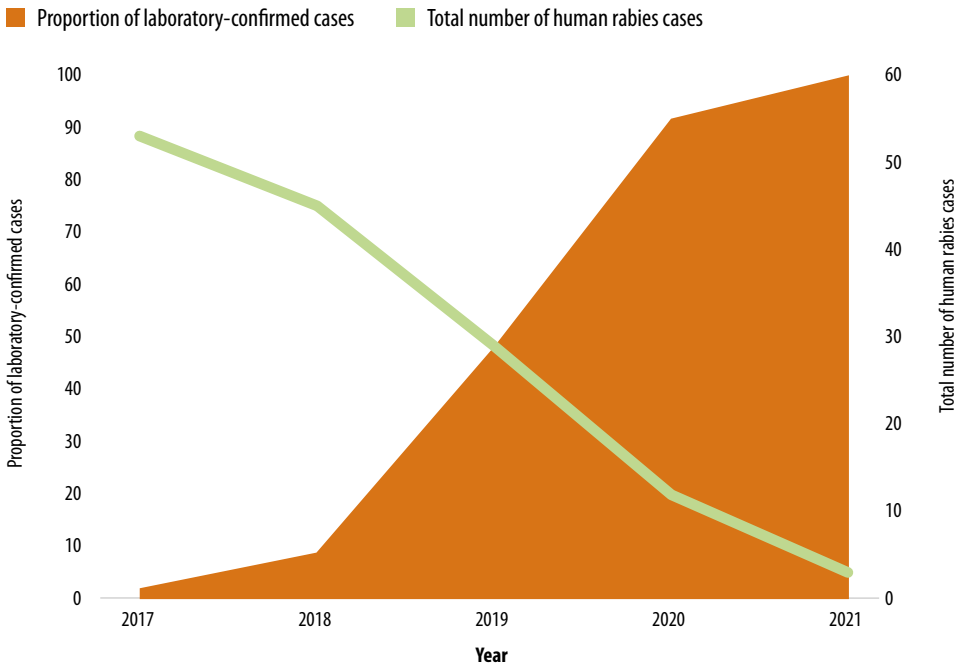
This section provides a brief description of each of the core analyses that are recommended on the diagnosis of rabies, and factors to take into consideration in their interpretation.

6.3.1 Proportion of laboratory-confirmed cases

Number of rabies cases that have been laboratory-confirmed / Total number of rabies cases × 100

Rabies cases can be diagnosed based on clinical examination and epidemiological exposure history as well as being laboratory-confirmed (Fig. 6). This indicator provides information on the diagnostic and laboratory capacity and can be used as an indicator for the testing rate in the country or in smaller administrative districts if disaggregation is available. Adequate testing is necessary to determine the true absence or presence of rabies in an area for better planning and canine-rabies free determinations.

Fig. 6. Proportion of laboratory-confirmed rabies cases compared with clinically-confirmed cases as a percentage. For reference, the total number of human rabies cases is displayed as well. The percentage of laboratory-confirmed cases increased over the past years consistently while the total case number decreased.



Source: WHO Global Health Observatory *(4)*.

6.3.2 Additional analysis

Rabies virus variant (if laboratory-confirmed)

Basic identification and classification of isolates can be used to identify the principal animal reservoir (e.g. bat, dog, fox, mongoose) and to infer the source of infection when a definitive history of exposure is lacking. It is needed to confirm canine-rabies freedom and to distinguish imported and locally acquired cases in rabies-free areas.

6.4 Management

This section provides a brief description of each of the core analyses that are recommended on treatment of rabies, and factors to take into consideration in their interpretation. Before interpreting the trends in treatment data, it is key to investigate issues of data quality. If you see any sudden changes in trends, first check data to make sure there has not been any error in data entry. Please refer to section 5 for recommended data quality indicators to guide interpretation.

These indicators are important for tracking bite treatment outcomes including PEP completion and adverse events after RIG administration or vaccinations.

6.4.1 Number of people receiving post-exposure prophylaxis

This gives an overview of the total number of people receiving at least one PEP vaccination dose and wound care during the reporting period. Data should be disaggregated at least by sex, age and WHO exposure category to identify trends in who is receiving treatment.

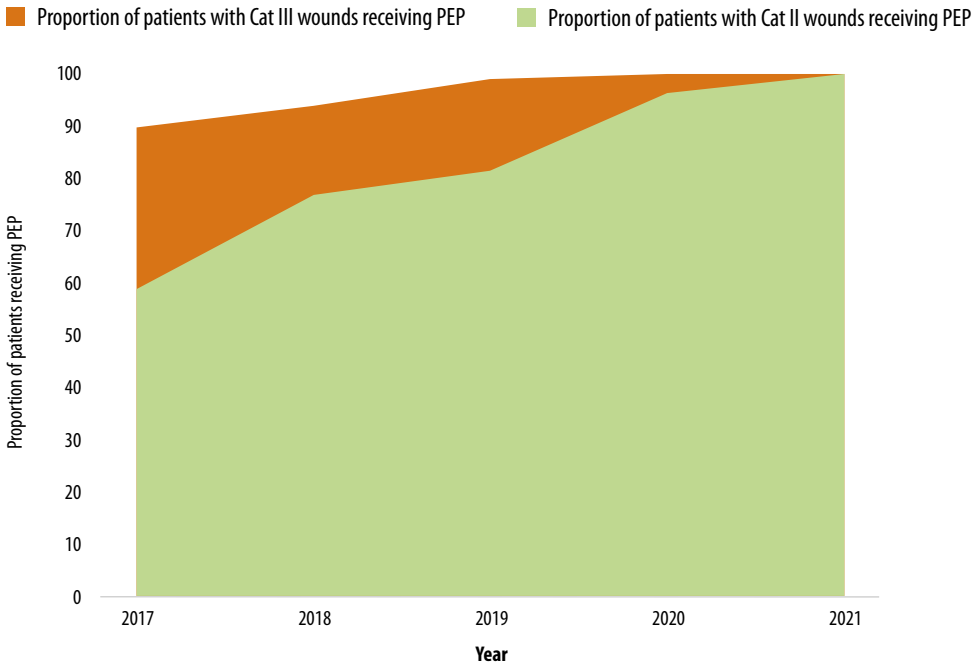
It can be used to assess the amount of vaccine that has been used in the country or if it is available in different regions or districts. This can be compared with the forecasted numbers and the procured vaccines. This assists in assessing the future need for PEP in an area.

6.4.2 Proportion of animal bite victims/exposed individuals receiving post-exposure prophylaxis

Total number of people receiving at least 1 PEP vaccination dose / Total number of humans bitten by an animal $\times$ *100*

This gives the proportion of people who have received PEP after being bitten by an animal (Fig. 7). This data should be disaggregated by at least sex, age and exposure category. If the percentage is low, countries need to investigate the reasons behind eligible bite victims not receiving PEP (e.g. unavailability of vaccine, lack of capacity or knowledge). This can be accomplished by conducting bite victim follow up surveys.

Fig. 7. **Proportion of patients receiving PEP disaggregated by wound category. In the beginning, especially treatment for category II wounds (scratches or abrasions without bleeding) was lacking. The number of eligible patients who received PEP increased over the years.**



Source: WHO Global Health Observatory *(4)*.

6.4.3 Additional analysis

Health-care facility catchment area

This can give an overview of the service delivery/accessibility to treatment. If patients need an unreasonable amount of time to reach the next health facility/anti-rabies clinic, a better distribution of services/decentralization should be considered. If the travel time to the health facility is too long, patients will be less likely to seek treatment or finish their PEP schedule and more people will be at risk.

Total number of bite victims/exposed individuals receiving rabies immunoglobulin (RIG) or rabies monoclonal antibodies (RmAbs)

This gives an overview of how many bite victims received RIG/RmAbs during the reporting period. This should be disaggregated by sex, age, species of the biting animal, WHO wound category and type of biological. It can be used to assess the eligibility of RIG/RmAbs administration and to get an overview of where RIG is administered if this information is included. This will also quantify RIG/RmAbs usage in the country to aid in future planning.

Proportion of bite victims/individuals with category III exposures eligible for RIG/RmAbs that receive RIG/RmAbs

Total number of people receiving RIG/RmAbs / Total number of patients eligible for RIG/RmAbs[1] $\times$ *100*

RIG is recommended for category III bite victims who have never received any rabies vaccination (an exception are immunocompromised persons, for whom RIG is recommended for category II and category III bite victims even if previously immunized). Data ideally are available disaggregated for sex, age and the species of the biting animal. Assessing the percentage of people who are eligible for RIG administration helps to assess its distribution and availability. It can identify a need for increased procurement and education of health-care workers on usage.

[1] Total number of patients with WHO wound category III exposures without prior rabies vaccination + Total number of immunocompromised patients with WHO wound category II exposures.

Proportion of bite victims/exposed individuals receiving PEP completing their vaccination schedule

Total number of people finishing their PEP schedule / Total number of people receiving at least 1 PEP vaccination dose × 100

Identifies the percentage of people who receive all scheduled vaccinations and assesses compliance in finishing their PEP schedule. It ideally is disaggregated by sex, age, WHO wound category, species of the biting animal and anatomical location of the bite. It can help to understand the service coverage of different areas. If the percentage is low and loss of follow up of patients is high, reasons need to be investigated (e.g. service delivery in the area, active outreach to patients, reminders, education). This can be accomplished through the use of bite victim follow up surveys.

Proportion of people showing adverse events to vaccination

Total number of people showing adverse events to rabies biologicals / Total number of people receiving rabies biologicals × 100

Identifies the occurrence of adverse events after receiving rabies biologicals. The information should disaggregate the suspected cause of reaction (eRIG, hRIG, RmAbs, vaccine) as well as sex and age. Type of adverse reaction can further be specified.

Proportion of people who receive PEP intradermally (ID)

Number of people who receive PEP ID / Number of people who receive PEP × 100

Identifies the percentage of people receiving ID in comparison to intramuscular (IM) regimens. Helps to identify cost-efficiency and potential need for education and training as ID vaccination schemes need less vaccine and visits while being as safe and efficient as IM vaccination if administered correctly.

Proportion of rabies biologicals used

Number of rabies biologicals used / number of rabies biologicals supplied to the national programme × 100

This compares the amount of rabies biologicals (disaggregated by RIG, RmAbs and vaccine) that has been supplied with the amount that was actually used, and helps to identify optimal distribution of biologicals available to reduce wastage.

6.5 Management of outcomes

This section provides a brief description of each of the core analyses that are recommended on management outcomes of rabies, and factors to take into consideration in their interpretation. Before interpreting the trends in treatment outcome data, it is key to investigate issues of data quality. If you see any sudden changes in trends, first check data to make sure there has not been any error in data entry. Please refer to section 5 for recommended data quality indicators to guide interpretation.

6.5.1 Analysis

Number of rabies cases having received rabies biologicals

Disaggregated by RIG and vaccination (including how many doses). Helps to assess the effectiveness of animal bite management and compliance. It further helps to detect potential treatment failures due to, for example, improper vaccine storage, vaccination techniques, vaccine quality, PEP completion, time to get treated.

Wound location and exposure category of rabies cases

This provides additional information on the way of transmission (e.g. higher risk of bites in highly innervated areas, multiple bites and deep wounds). Information helps to adjust risk-assessment schedules for patient treatment.

Treatment outcomes (number survivors, number of deaths)

If any treatment of rabies cases is performed, outcomes should be documented disaggregated by sex, age, wound category and previous application of rabies biologicals.

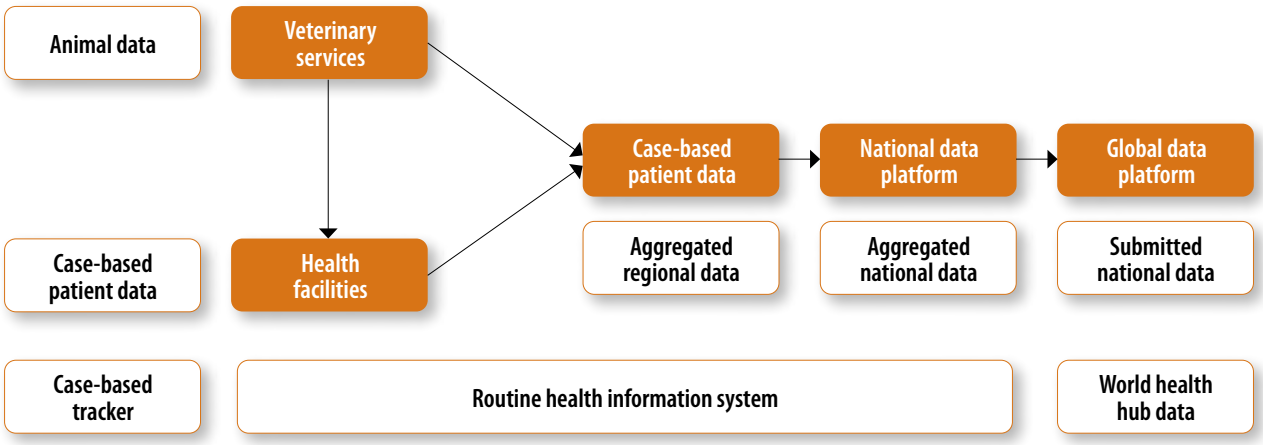
7. Global reporting and validation of "zero human deaths"

Data collection and sharing are crucial pillars for ending the burden of rabies. Improving the quality of global rabies data is key. The collection, analysis, publication and dissemination of data form a core part of WHO's mandate.[1] National focal points are responsible for reporting their aggregated rabies surveillance data to WHO. More information on how the data can be reported as well as reporting templates are available on WHO's website *(9)*.

The information collected with national routine health information systems feed into the global rabies indicators as defined in the road map *(1)* and the companion monitoring and evaluation framework *(10)*.

WHO developed the World Health Data Hub as a comprehensive digital platform for global health data (Fig. 8). It provides end-to-end solutions to collect, store, analyse and share data that are timely, reliable and actionable. Data on cases of human rabies reported by Members States to WHO are made publicly accessible through the NTD road map tracker *(11)*, the WHO Global Health Observatory *(4)* and its related platforms on the World Health Data Hub. This will regularly be updated to depict the global progress of the 155 countries (92%) targeted to achieve "Zero by 30" (zero human dog-transmitted rabies deaths by 2030).

Fig. 8. **Data flow from local case-based recording in health facilities to the global level**



Additionally, Table 3 presents programmatic level indicators which help measure country-level progress towards the goal of eliminating rabies as a public health problem.

[1] For more information, see WHO data policy *(12)*.

Table 3. Global level rabies indicators to 2030

Global indicators	Definition
Number of countries having achieved zero human deaths from dog-mediated rabies	Countries reporting zero deaths due to dog-transmitted rabies by the end of 2030. "Zero human deaths" is defined as interruption of dog to human rabies transmission and no human deaths. Implementation of procedures for validation of elimination as a public health problem as outlined in Technical Report Series (TRS 1012).
Number of countries having reduced mortality due to dog-transmitted human rabies by 50%	Number of countries having reduced the number of human cases due to dog-transmitted rabies by 50% since the first NTD road map (2012).
Number of countries having reached 70% vaccination coverage of dogs in high-risk areas	Number of countries controlling rabies at source by implementing mass dog vaccination programmes and targeting high-risk areas of dog rabies transmission to break the cycle. Vaccination coverage of dogs (both unowned and owned) in high-risk areas is measured through community-based surveys by veterinary public health services. The health ministry should make sure to monitor and report these data to WHO on annual basis.

However, sharing national rabies surveillance data globally goes beyond measuring country progress towards global targets and indicators as described in Table 4. There are many more reasons why countries should submit their data. The main advantages are described below and are also available as a flyer (Fig. 9 and 10). Suggested further resources are provided in Annex 6.

Table 4. Main reasons for countries sharing national data with WHO

Positive effect of data sharing	How the data are used
Increase knowledge, create impactful advocacy and raise rabies awareness	National rabies data are crucial to understand the burden of rabies globally and to create more comprehensive estimates of mortality, exposures and health interventions. This will support outreach and create more powerful and, for the audience, relevant messages to increase awareness and attention to end the suffering from rabies.
Inform resourcing needs	Global data can be used to assess the needs for innovation and resources (e.g. forecasting of rabies biologicals or diagnostics both for human and animals).
Inform policy decisions	Country data provide for better-informed activities to establish global health guidelines, international norms and standards.
Empower national decision-making	Access to data from a wide range of countries improves public health and resource allocation by empowering Member States to make better, evidence-based decisions for preventing and controlling rabies.
Enable collaboration	Rabies data enable collaboration between human and animal sectors towards a One Health approach including transboundary and regional initiatives.
Advance scientific understanding	Accurate data allow for analysis and hypothesis testing by multiple groups of researchers, promote more complete and reliable systematic reviews and meta-analyses.
Improve transparency and accountability	Rabies data maximize transparency and accountability in tracking global progress of rabies elimination programmes and in the conduct and funding of scientific investigations and research.
Meet WHO Member State requirements	The collection, analysis, publication and dissemination of data form a core part of WHO's mandate. Your data permit WHO and other partners to help establish and curate national databases – to eliminate errors in data, and to support the correct interpretation of data, including by repeated independent validation.
Measure progress towards targets and goals	Data are used to understand country progress against country targets and towards the global goal of reaching zero dog-transmitted human rabies deaths by 2030.
Validate your success	Only with data can "rabies free" areas be validated and elimination proven.

Fig. 9. Sharing rabies data: commit to submit

Source: Sharing rabies data: commit to submit (13).

Fig. 10. Validate your success

Source: Validation of elimination of dog-mediated human rabies as a public health problem (14).

Validation offers countries the opportunity to demonstrate their success and to show that ending infectious diseases is possible. A country previously endemic for rabies may apply for validation as having eliminated dog-mediated rabies as a public health problem ("Zero human deaths"), as described in the WHO expert consultation on rabies *(15)*, if it fulfills the following conditions:

- no human death from dog-mediated rabies for at least 24 months;
- operation and maintenance of an adequate rabies surveillance system as described in the WHO technical report series (TRS 1012); and
- demonstrating effective implementation of a rabies control programme in human and animal populations.

A draft template dossier explaining core requirements is described in Annex 14 of the expert consultation; an editable version can be downloaded *(16)*. Each section of the template dossier has to be addressed and supported with a presentation of respective programme data.

Investing in rabies elimination saves lives and strengthens both human and veterinary health systems. A collaborative response, through rabies elimination programmes, contributes to disease prevention and preparedness and is a model for One Health collaboration that can make Zero by 30 a reality.

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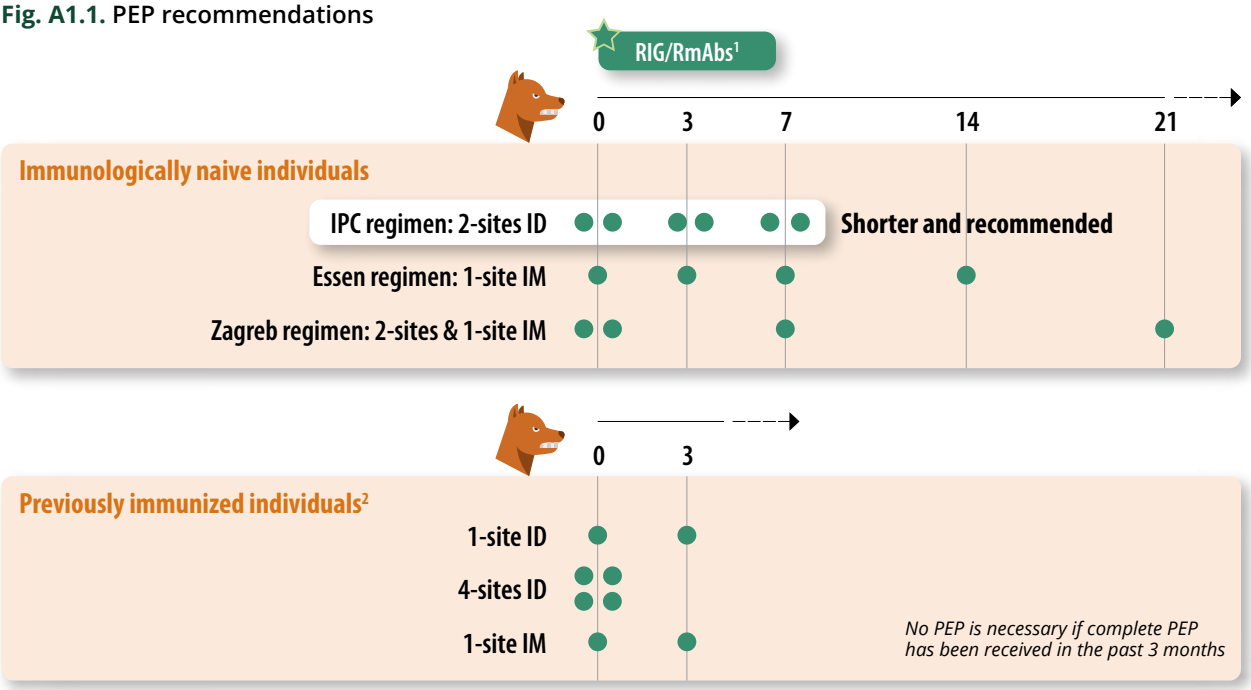
Annex 1. **Recommendations on post-exposure prophylaxis**

Recommendations on post-exposure prophylaxis (PEP) depend on the WHO wound category (Category II or III) as well as on the rabies vaccination history of the patient (Fig. A1.1). Thorough wound washing with copious amounts of water and soap for 15 min should always be performed. No PEP is required for Category I (no exposure).

Fig. A1.1. PEP recommendations



ID: intradermal; IM: intramuscular; PEP: post-exposure prophylaxis; RIG: rabies immunoglobulin; RmAbs: rabies monoclonal antibodies.

[1] RIG or RmAbs should be administered as soon as possible. If a patient has ever received any rabies vaccination (PEP or pre-exposure prophylaxis) more than 7 days prior to a Category III exposure, RIG must not be given (exceptions exist for immunocompromised patients as illustrated in Annex 2).

[2] If full PEP < 3 months ago: no PEP needed.

Source: Rabies vaccines: WHO position paper – April 2018. Geneva: World Health Organization; 2018 (https://iris.who.int/handle/10665/272372).

PEP recommendations on the basis of surveillance capacity are provided in Table A1.1.

Table A1.1. Recommendations for PEP on the basis of surveillance capacity

Rabies surveillance programme	Programme description	When to initiate PEP	When to discontinue PEP
No routine surveillance	No trained professionals capable of assessing animals for rabies. No laboratory capacity for timely testing of samples.	Initiate PEP immediately	Do not discontinue PEP, unless a trained professional has confirmed that the animal is healthy 10 days after the bite.
Limited surveillance	Trained professionals capable of assessing animals for rabies are available in some communities. Laboratory capacity exists, but testing and reporting may be delayed.	Initiate PEP immediately	Do not discontinue PEP, unless tests at a qualified laboratory give negative results; or A trained professional has confirmed that the animal is healthy 10 days after the bite.
Advanced surveillance (i.e. integrated bite case management)	Trained professionals capable of assessing animals for rabies are consistently and reliably available in the community in which the exposure occurred. Laboratory capacity exists and can reliably test samples and report results within several days of the bite.	Initiate PEP immediately for bites to the head, neck, other highly innervated sites, multiple or deep wounds and for bites to children.[a] When the animal is available and determined by a trained professional to present a low risk, PEP may be delayed.	Do not discontinue PEP, unless tests at a qualified laboratory give negative results; or A trained professional has confirmed that the animal is healthy 10 days after the bite.

[a] Given their small stature and higher risk of severe exposure, children should receive PEP immediately.

Source: WHO expert consultation on rabies: third report. Geneva: World Health Organization; 2018 (https://iris.who.int/handle/10665/272364).

Annex 2. **Decision tree for wound risk assessment in clinical settings**

Wound risk assessment

Animal: mammal? — N → Species not susceptible for rabies: no PEP indicated, wash exposed areas with soap and water

Y ↓

Animal: rodent? — Y → Rodents rarely transmit rabies: no PEP indicated, wash exposed areas with soap and water

N ↓

Mucous membrane: contaminated with saliva? — N → Skin: broken? — N → Category I

Skin: broken? — Y ↓ Animal: bat?

Mucous membrane: contaminated with saliva? — Y → Animal: bat? — Y → Category III

Animal: bat? — N ↓ Wound: bleeding?

Wound: bleeding? — Y → Category III

Wound: bleeding? — N → Category II

ARV = Anti Rabies Vaccine
PEP = Post-Exposure Prophylaxis
RIG = Rabies Immunoglobulin
RmAbs = Rabies Monoclonal Antibodies
Y = Yes, N = No

Category I

WHO wound category I

Animal licks on intact skin, touching or feeding the animal

No PEP indicated, wash exposed areas with soap and water

Category II

WHO wound category II

Minor scratches or abrasions, nibbling of uncovered skin

Patient: immunocompromised? — Y → Wound washing[1], RIG/RmAbs and ARV immediately[2]

N ↓

Patient: ever received any rabies vaccination?[3] — Y → Patient: completed PEP less than 3 months ago? — Y → No PEP indicated, wash exposed areas with soap and water

Patient: ever received any rabies vaccination?[3] — N → Wound washing[1] and ARV

Patient: completed PEP less than 3 months ago? — N → Wound washing[1] and ARV[4]

Wound washing[1] and ARV → Proceed with PEP Risk Assessment[6]

Wound washing[1] and ARV[4] → Proceed with PEP Risk Assessment[6]

Category III

WHO wound category III

Transdermal bites or scratches, licks on broken skin, contamination of mucous membrane with saliva, exposure to a bat

Patient: immunocompromised? — Y → Wound washing[1], RIG/RmAbs and ARV immediately[2]

N ↓

Patient: ever received any rabies vaccination?[3] — Y → Patient: completed PEP less than 3 months ago? — Y → No PEP indicated, wash exposed areas with soap and water

Patient: ever received any rabies vaccination?[3] — N → RIG/RmABs availability: limited?

Patient: completed PEP less than 3 months ago? — N → Wound washing[1] and ARV[4]

RIG/RmABs availability: limited? — Y → Patient: high-risk?[5] — Y → Wound washing[1], RIG/RmAbs and ARV

RIG/RmABs availability: limited? — N → Wound washing[1], RIG/RmAbs and ARV

Patient: high-risk?[5] — N → Wound washing[1] and ARV

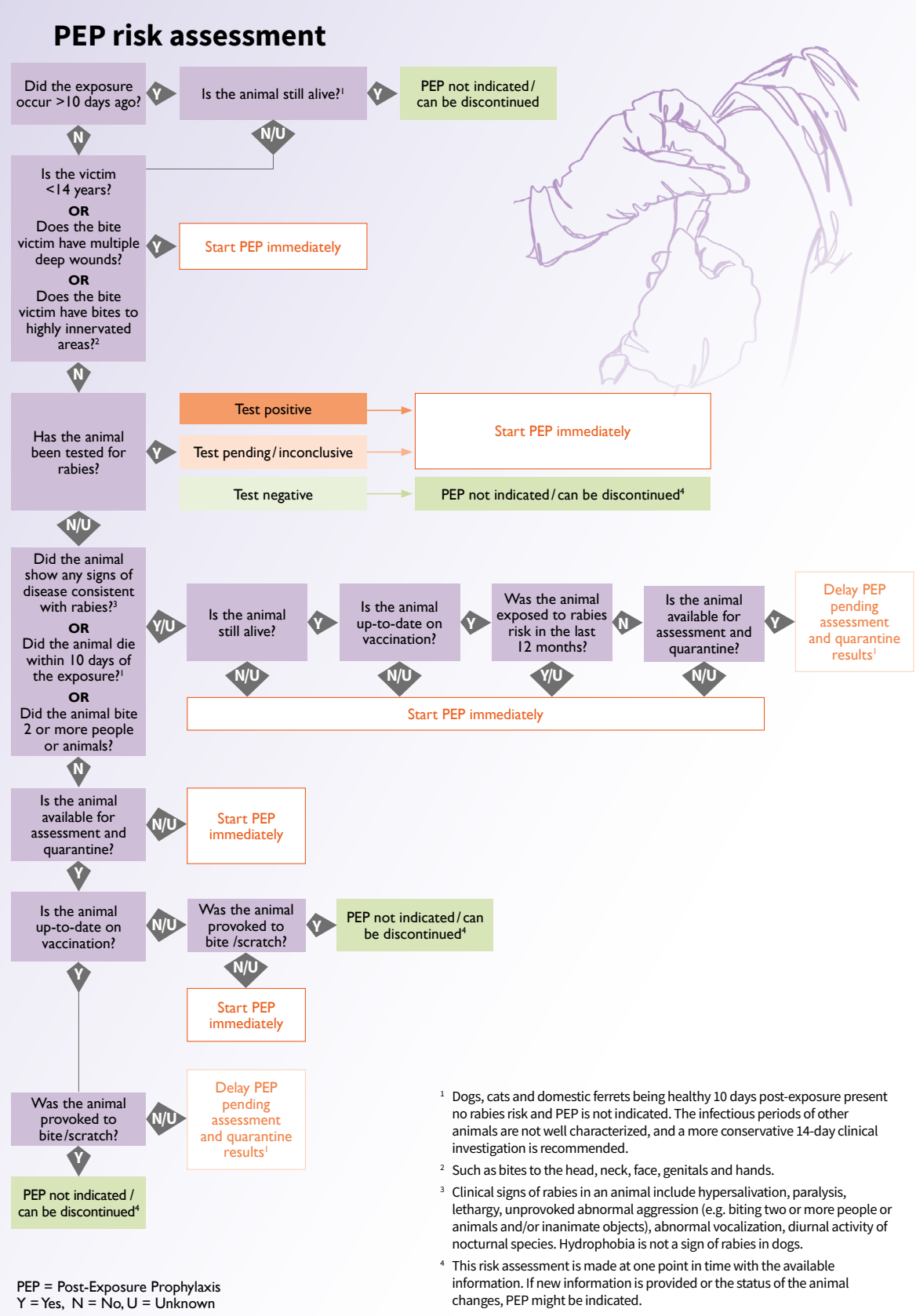
Wound washing[1], RIG/RmAbs and ARV → Proceed with PEP Risk Assessment[6]

Wound washing[1] and ARV → Proceed with PEP Risk Assessment[6]

Wound washing[1], RIG/RmAbs and ARV → Proceed with PEP Risk Assessment[6]

[1] Wash the wound thoroughly with copious amounts of water and soap for 15 min and apply an antiseptic. This is a life-saving practice especially for immunocompromised patients.
[2] When feasible, the RABV neutralizing antibody response should be determined 2–4 weeks after vaccination to assess whether an additional dose of vaccine is required.
[3] This includes Pre-Exposure Prophylaxis (PrEP), previous PEP, or patients beyond the 7th day of PEP.
[4] If PrEP or previous PEP was received: shorten the current vaccination schedule accordingly.
[5] Patient with multiple bites, deep wounds, bites to highly innervated parts of the body (such as head, neck, face, genitals and hands), severe immunodeficiency, bites from an animal with probable (clinically) or confirmed (laboratory) rabies, exposure to a bat (bite, scratch or exposure of mucous membrane).
[6] If your programme does not have adequate surveillance to assess the biting animal (as recommended in the PEP Risk Assessment Decision Tree), PEP should be initiated immediately.

Annex 3. **Decision tree for post-exposure prophylaxis risk assessment**



[1] Dogs, cats and domestic ferrets being healthy 10 days post-exposure present no rabies risk and PEP is not indicated. The infectious periods of other animals are not well characterized, and a more conservative 14-day clinical investigation is recommended.

[2] Such as bites to the head, neck, face, genitals and hands.

[3] Clinical signs of rabies in an animal include hypersalivation, paralysis, lethargy, unprovoked abnormal aggression (e.g. biting two or more people or animals and/or inanimate objects), abnormal vocalization, diurnal activity of nocturnal species. Hydrophobia is not a sign of rabies in dogs.

[4] This risk assessment is made at one point in time with the available information. If new information is provided or the status of the animal changes, PEP might be indicated.

Annex 4. **Glossary for case-based reporting form**

Elements marked with an asterisk (*) can provide additional information.

Enrolment	
Registration date (dd/mm/yyyy)	Write the date of today (e.g. the date the patient is visiting)
Date of exposure (dd/mm/yyyy)	Write the date of exposure to the animal.
First name of patient	Write the first name of the patient.
Last name of patient	Write the family/last name of the patient.
Date of birth (dd/mm/yyyy)	Write the date of birth of the patient.
Sex	Select **Female** or **Male**
Address*	*Write the location where the patient comes from including the name of the town or village, name of district, name of the region, province or state. If the patient is from a country different from where they are seeking treatment, include that as well.* *Note: the name of the administrative divisions may vary from one country to another.*
Contact telephone number	Write the telephone number for the patient.
Email address	Write the email address for the patient.
Rabies case ID#	Automatically generated identification number given to the patient.

Case assessment	
Date of assessment (dd/mm/yyyy)	Write the date that the patient was seen in the health facility.
Current health status	Select the current health status of the patient **Healthy, Ill** or **Deceased** Ill – consistent with rabies is a person presenting with an acute, progressive neurological syndrome (i.e. encephalitis) showing hyperactivity (furious rabies) or paralysis (paralytic rabies) and may include any of the following signs: aerophobia, hydrophobia, paraesthesia or localized pain, dysphagia, localized weakness, nausea or vomiting. This is a **suspected rabies case**. Rabies vaccine should not be administered to persons suspected to have clinical rabies, as response to the vaccine could interfere with interpretation of antemortem diagnostic testing and may expedite the clinical process, resulting in more rapid disease progression.
Current immune status	As there are special recommendations for immunodeficient patients, it is important to consider the current immune status. Select **Healthy/Normal** or **Immunodeficient** For immunocompromised individuals (such as HIV-infected persons who are not receiving antiretroviral therapy or who are receiving antiretroviral therapy but do not meet minimum CD4 cell-count criteria) with WHO category II and III rabies virus exposure, the following is recommended: thorough washing of the wound should be emphasized; a full course of rabies vaccine should be administered, plus RIG in all cases, even if previously immunized.
Patient age (years)	Automatically calculated based on the patient's birthday and current date.

Patient weight (kg)	Write the body weight (bw) of the patient in kilograms (kg) at the time of the visit. This weight can be used to calculate RIG/RmAb dosage (if indicated). Maximum dosage is 20 IU (hRIG) and 40 IU (eRIG) per kg bw and 3.33 IU for single mAb and 40 IU for cocktail RmAbs per kg bw.
History of rabies vaccination	As previous rabies vaccination will influence the treatment recommendation, ask the patient about their vaccination history and select **Pre-Exposure Prophylaxis; Post-Exposure Prophylaxis; Discontinued, but received > 2 doses; Discontinued, but received < 2 doses** or **Never vaccinated against rabies**. No PEP is necessary if complete PEP has been received in the past 3 months. Having already received at least 2 doses of rabies vaccine at some point in life counts as PrEP. Previously vaccinated patients are subject to a shorter vaccination schedule and RIG or RmAbs are not indicated.
Date of previous vaccination*	*Only necessary for persons with a history of vaccination.* *Write the date of vaccination.* *If the exact date is unknown, enter how many years, months or days ago the last vaccine administration took place.*
Details of previous vaccination history*	*Only necessary for persons with a history of vaccination.* *Write the name of the used vaccine and location of administration if known.*
History of adverse reactions to rabies vaccine (or any other common vaccines)?*	This is to understand the risk of possible adverse events after vaccination. Patients with known reactions to rabies or other common vaccines should be monitored carefully after vaccination (if applicable). Select **Yes, severe reactions; Yes, minor reactions; No** or **Unsure** If the patient had a mild local reaction (e.g. pain, redness or swelling), they can continue PEP using the same brand of rabies vaccine. If they had a severe local or systemic reaction, another type of rabies vaccine should be used, if available (e.g. purified chick embryo cell vaccine, purified vero cell rabies vaccine).
Any symptoms consistent with rabies?	*Only relevant if current health status is "Ill" or "Deceased".* *If patient is showing symptoms consistent with rabies, document all known symptoms.* *A suspected clinical case of rabies in humans is defined as: an acute neurological syndrome (i.e. encephalitis) dominated by forms of hyperactivity (furious rabies) or paralytic syndromes (paralytic rabies) progressing towards coma and death.*
Was any relevant diagnostic testing performed?* ***(Disease/Test/Date/Result/Comment)***	*Only relevant if current health status is "Ill" or "Deceased".* *Describe here the cause of encephalitis if known, relevant tests that have been performed, the date and the result of the test including comments if relevant (e.g. Encephalitis; Rabies; Mosquito-borne encephalitis; Herpes simplex virus; Zoster encephalitis; Enterovirus; Measles virus; Bacterial meningitis; Malaria; Toxoplasmosis).*
Was the patient diagnosed with encephalitis showing hyperactivity (furious rabies) or paralysis (paralytic rabies) and died within 7–10 days from onset of clinical signs?	*Only relevant if current health status is "Deceased".* *A patient showing signs of rabies progresses towards coma and death, usually due to cardiac or respiratory failure, typically within 7–10 days of the first sign if no intensive care is instituted.* *This is considered a **suspected rabies case**. If the patient was in contact with a suspected/probable or confirmed rabid animal it is a **probable rabies case**. Both cases can be considered **clinically confirmed**.*
If the patient was tested positive for rabies – rabies virus variant*	*Only relevant if patient has been tested positive for rabies.* *Specify the rabies virus variant of the positive tested patient.* *Select **Dog; Bat; Terrestrial carnivore; Other** or **Unknown**.*

Exposure wound assessment	
Date of exposure (dd/mm/yyyy)	Write the date of exposure to the animal. If already entered during the enrolment phase, this field will be pre-populated. Days since exposure should be calculated.
Anatomical location of exposure	Select all anatomical locations that were exposed to: ***Head; Neck; Trunk; Upper limb; Hands; Lower limb; Genitalia*** or ***Other***.
Number of wounds	Write the number of wounds the patient has as a result of the animal exposure.
WHO wound category (I, II, III)	Specify the exposure by selecting one of the wound categories I–III as classified by WHO. **Category I**: The skin is still intact: animal licks on intact skin, touching or feeding animals. This is not an exposure. **Category II**: The skin is broken but there is no bleeding (minor scratches, nibbling of uncovered skin). **Category III**: The skin is broken and the wound is bleeding, mucous membrane or broken skin has been contaminated with saliva or animal contacts with wild animals including bats has occurred. This is considered a severe exposure. If multiple exposure categories apply for different anatomical locations, use the most severe category.
Was the wound treated prior to coming (e.g. at home or from another facility, traditional healer, medical practitioner)?	Ask the patient, in a polite and non-intimidating way, if the wound was already treated prior seeking medical consultation (e.g. wound has been washed appropriately, home treatment, treatment by a traditional healer). If no appropriate wound treatment has been conducted yet, the wound should be washed and flushed for approximately 15 min, with soap or detergent and copious amounts of water. Where available, an iodine-containing, or similar viricidal, topical preparation should be applied to the wound. Wound washing is the most important first aid.

Exposure risk assessment	
What type of animal was the patient exposed to?	Write down the type of animal the patient was exposed to: Select **Bird; Fish; Reptile; Dog; Cat; Bat; Livestock; Rodent; Other Mammalian Wildlife** or **Other (specify)**. If patient was exposed to a **bird, fish** or **reptile**, PEP is not indicated as these species are not susceptible to rabies. If the patient was exposed to a rodent, PEP is not indicated, as rodents rarely transmit rabies.
Animal health status	Healthy; Sick; Animal is deceased; Unknown *If the animal is an unknown/stray dog, then a more conservative approach should be used to decide if the animal was truly healthy. If there is not ample evidence that the animal was healthy at the time of the bite, then "Unknown" should be the default choice.*
Any signs consistent with rabies?	*Only applicable if the animal health status is "Animal is deceased" or "Sick".* *Ask the patient if the animal has shown any signs of disease.* *An animal that presents any of the following signs –hypersalivation, paralysis, lethargy, unprovoked abnormal aggression, abnormal vocalization, activity during the day of night active animal species – is considered a* ***suspected rabid animal****.* *Select* ***Yes*** *or* ***No*** *and indicate what signs the animal was exhibiting.*
Has the animal been tested for rabies?	Ask the patient if they know if the animal they were exposed to has been tested for rabies. Select **No / Unknown – Animal NOT currently undergoing testing; Yes – Test results PENDING; Yes – Test results NEGATIVE; Yes – Test results POSITIVE** or **Yes – Test results INDETERMINATE**. If the animal has been tested negative, PEP may not be indicated.

Is the animal available for veterinary assessment, quarantine or diagnostic testing?	Ask the patient if the animal they were exposed to is available for investigation. Select **Yes – Bite victim's own animal; Yes – Owner is known; Yes – Other; No – Owner will not comply with veterinary assessment or quarantine; No – No veterinary staff or quarantine available; No, animal has been killed by owner/community or has died** or **No, Location of the animal is unknown**. If the animal is available for assessment, respective veterinary health professionals/workers should be contacted and informed. Results of the investigation should be shared with the health facility. PEP can be discontinued if the suspected rabid animal is proven to be free of rabies by appropriate laboratory examination; or, in the case of domestic dogs, cats or ferrets, the animal remains healthy throughout a 10-day observation period starting from the date of the bite.
Does the animal have a history of rabies vaccination?	Only if the animal is the bite victim's own animal or the owner is known. Select **Yes – Current on vaccination; Yes – Overdue for vaccination; No** or **Unknown**. Animal vaccination should not be used to determine PEP but can be used to determine if PEP could be safely delayed and should ONLY be used in countries with advance surveillance capacity. If the vaccination status of the animal is uncertain, "no or unknown" should be selected. If possible, proof of vaccination, such as a vaccination card or license, should be requested.
Was the animal provoked to bite/ scratch?	Ask the patient, in a polite and non-intimidating way, if they could have provoked the animal. A good way to ask indirectly is to have the patient recall how the bite/scratch/exposure happened and what the animal as well as the patient was doing. Select **Yes – Victim disturbed the animal; No – Animal bite/scratched without provocation** or **Unknown**. Bites inflicted on a person attempting to feed or handle an apparently healthy animal should generally be regarded as provoked.
Was the animal exposed to rabies in the last 12 months? (typically "unknown" for animals allowed to roam freely)	Select **Yes; No** or **Unknown** (typically for free-roaming dogs)
Where did the bite occur (city/ neighbourhood)	Use the map feature to locate the spot where the bite occurred. This information can be used to identify risk areas or new entrances in rabies-free areas. Information can be shared with local authorities to raise awareness about potential risks.
How many other people were exposed to this animal? (enter zero if unknown)	Write down the number of other people (not including the patient) who were bitten or scratched by the animal, including their contact details if available. Animals that have bitten more people have a higher risk of being infected with rabies. A responsible authority should be contacted to contact other potential victims and information about PEP should be shared with them.
How many other animals were exposed to this animal? (enter zero if unknown)	Write down the number of other animals that were bitten or scratched by the animal, including the contact details of the owners if available. Animals that have bitten other animals have a higher risk of being infected with rabies. The responsible animal health authority should be contacted to contact animal owners and inform them about the situation.

Assessment determination	
Species risk determination	Animal susceptibility to rabies, see decision tree in Annex 2
Wound Assessment Determination	See decision tree in Annex 2
RIG recommendation	See decision tree in Annex 2
ARV recommendation	See decision tree in Annex 2

Risk assessment determination	See decision tree in Annex 3 The first dose of rabies vaccine should be administered immediately after a potential exposure. Vaccine should always be administered when a category III exposure is recognized, even weeks or months after the contact. When vaccine supply is limited, suspected and probable exposures that occurred more than 12 months previously may be considered of lower priority for vaccination, as clinical rabies rarely appears more than 12 months after exposure.

Clinical visit	
Date of visit (dd/mm/yyyy)	Write the date of the clinic visit when the patient received rabies vaccine or immunoglobulin.
Are animal test results available?*	Rabies test results may not always be available when the patient starts PEP. If the animal tests negative at any point during the vaccine series, rabies vaccine can be safely discontinued. Alternatively, if the animal tests positive, it becomes imperative that the patient completes the vaccine series. Yes; No; Unknown
Select the biologicals that are given to the patient	Select if rabies immunoglobulin (RIG) or anti-rabies vaccine (ARV) were administered during the visit. RIG is recommended for severe exposures (Category III). It is not recommended if you have ever received rabies vaccines (with the first dose received more than 7 days ago), as circulating virus-neutralizing antibodies will interfere with the passive immunization. When RIG amount is limited, the following cases should be prioritized: Category III exposures with multiple bites, deep wounds, bites in highly innervated body parts (head, neck hands); patients with severe immunodeficiency; and cases from confirmed/probable positive animals (bites/scratches or exposure of mucous membrane caused by a bat). Wounds that require suturing should be sutured loosely and only after RIG infiltration into the wound, in addition to proper wound care and tetanus boosters, if applicable.
Select the kind of RIG or RmAb administered	Only applicable if RIG is given. Select the kind of biological administered (equine or human rabies immunoglobulin or single or cocktail RmAbs)
RIG/RmAb amount recommended	Only applicable if RIG is given. Calculated with the weight of the patient. Maximum dosage is 20 IU (hRIG) and 40 IU (eRIG) per kg bw and 3.33 IU for single mAb and 40 IU for cocktail RmAbs per kg bw.
Date RIG/RmAb administered *(dd/mm/yyyy)*	Only applicable if RIG is given. Usually, the date of the clinic visit.
RIG/RmAb product name/lot number*	Only applicable if RIG is given. Note the product name and the lot number.
Anatomical location of RIG/RmAb administration	If RIG has been administered, write down the anatomical location.
Rabies vaccination series selected	Select the vaccination schedule that has been selected for the patient. The choice of vaccination schedule should consider feasibility (e.g. cost, number of doses, time and compliance), and may also depend on clinical settings and patient preference. WHO recommends a shortened 1-week intradermal vaccination regimen (2-site vaccination on d0, d3 and d7).

Anatomical location of vaccine administration	Describe how the vaccine has been administered (e.g. IM, one-sited deltoid area). WHO recommends administration of rabies vaccines either intradermally or intramuscularly. Depending on the chosen schedule, rabies vaccination can require the administration of multiple injections during the same visit. If 2 or more doses of rabies vaccine are administered at the same visit, they should be injected in different sites/limbs, and always administered away from the site of RIG. Vaccines should not be injected in the gluteal region.
Date vaccine administered (dd/mm/yyyy)	If rabies vaccine has been administered, write down the date. This is usually the date today.
Vaccine product name/lot number*	If ARV has been administered, note the product name and the lot number.
Did the patient have any adverse reactions to vaccine or RIG/RmAbs? If yes, describe	Select **Yes (please specify); No** or **Unknown**. Describe any adverse events that occurred after vaccination.
Site deviation	If vaccine or RIG is administered inappropriately, a notice will display indicating a possible deviation. Vaccine should not be administered in the glute or in the same anatomical site as RIG. This would be a common cause of a possible deviation. In these instances, the responsible public health authority should be contacted for additional advice.
Next vaccination date (dd/mm/yyyy)	The date of the next vaccination will populate based on the date of visit and selected vaccination schedule. Ensure that it is communicated to the patient that they should try to stick to the vaccination schedule as closely as possible.
Schedule deviation	All attempts should be made to main the selected PEP schedule. While minor delays of less than a week are generally acceptable, the effects of larger deviations are less known.
Clinical visit comment	Enter any further information/comments.

Annex 5. **Animal-based indicators to be collected at the national level**

As rabies is transmitted by animals (mostly dogs), surveillance systems must include animal data. Primary history of biting animals should be collected, if available, but further information on rabies incidence in animals, the dog population and the proportion of vaccinated dogs is crucial.

This information is often collected by veterinary health professionals and should be shared with human health counterparts. In systems where both the human health and veterinary sectors both have surveillance systems for animal rabies, a data sharing agreement should be established to ensure that all animal rabies incidence data are maintained in one national data repository. Animal rabies should be considered an immediately reportable event, triggering data sharing with the appropriate sectors and follow-up on any human or animal exposure (see Table A5.1).

Table A5.1. Rabies animal-based indicators collected by WHO

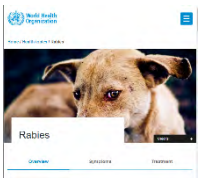
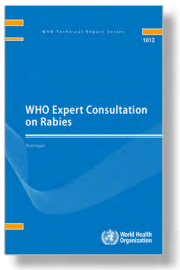
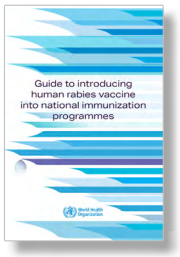
Indicator	Description	Disaggregation	Comments
Number of animal rabies cases reported	Number of animal rabies cases during the reporting period (i.e. in one year)	• Type (dog, cat, bat, wildlife, livestock, unknown) • Diagnosis (clinical diagnosis, laboratory diagnosis)	
Dog population	Estimation of dog population in the country (including method of estimation and proportion of free-roaming dogs)	• Ownership status (unowned, owned)	
Proportion of vaccinated dogs	Proportion of vaccinated susceptible dogs during the reporting period	• Ownership status (unowned, owned)	Dog vaccination coverage

Additional indicators:

- Total numbers of animals tested
  Disaggregated by animal
- Proportion of positive tested animals
  Disaggregated by animal
  Number of positive tested animals / Total number of all tested animals

Further information on animal rabies is accessible from the WOAH homepage (https://www.woah.org/en/disease/rabies/).

Annex 6. **Further resources**

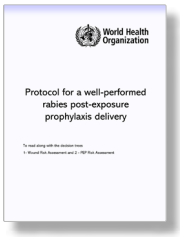
Rabies-specific WHO resources (IRIS handle/web link)		
	**WHO homepage on rabies** (https://www.who.int/health-topics/rabies#tab=tab_1)	The landing page for all rabies-relevant topics from WHO.
	Zero by 30: the global strategic plan to end human deaths from dog-mediated rabies by 2030 (https://iris.who.int/handle/10665/272756)	In 2015, the world called for action by setting a goal of zero human dog-mediated rabies deaths by 2030, worldwide. The United Against Rabies collaboration leverages existing tools and expertise in a coordinated way to empower, engage and enable countries to save human lives from this preventable disease. The global strategic plan puts countries at the centre, with renewed international support to act.
 French, Spanish	**WHO Expert Consultation on rabies: third report** (https://iris.who.int/handle/10665/272364)	This report provides updated guidance based on evidence and programmatic experience on the multiple facets of rabies prevention, control and elimination. Key updates include: (i) surveillance strategies, including cross-sectoral linking of systems and suitable diagnostics; (ii) the latest recommendations on human and animal immunization; (iii) palliative care in low-resource settings; (iv) risk assessment to guide management of bite victims; and (v) a proposed process for validation and verification of countries reaching zero human deaths from rabies.
	**Guide to introducing human rabies vaccine into national immunization programmes** (https://iris.who.int/handle/10665/360978)	The new guide introduces the human rabies vaccine and aims therefore to support national immunization programme managers as well as others responsible for implementing vaccination programmes. This will ensure that: policy discussions are properly informed and operational planning can be carried out for the introduction or expansion of rabies PEP into a national immunization programme; and considerations specific to rabies PEP are highlighted, securing proper integration with existing systems, including for implementation, and monitoring and evaluation. The guide outlines six steps for countries to follow. For each step, WHO provides technical information and guidance specific to rabies. Where relevant, it also provides examples and links to further resources.

Arabic, Russian

Rabies vaccines: WHO position paper – April 2018

(https://iris.who.int/handle/10665/272372)

This position paper presents new evidence in the field of rabies and the use of rabies vaccines, focusing on programmatic feasibility, simplification of vaccination schedules and improved cost–effectiveness. The recommendations concern the two main immunization strategies, namely vaccination for post-exposure prophylaxis (including the use of rabies immunoglobulins) and vaccination for pre-exposure prophylaxis.



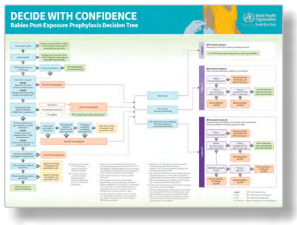
French

Protocol for a well-performed rabies PEP delivery

(https://doi.org/10.2471/B09018)

Here we offer health-care providers up-to-date, simple, exhaustive, visual guidance on how to perform PEP steps correctly.

The PEP protocol involves: (i) washing the wound with water and soap for 15 min; (ii) assessing the need for rabies biologicals based on specific criteria; (iii) administering, if required, rabies immunoglobulin or monoclonal antibodies deep in and around all wounds; (iv) starting, if necessary, the WHO-recommended intradermal 1-week vaccination regimen; and (v) informing patients adequately throughout the PEP process to increase compliance and avoid dangerous misconceptions about animal bite treatment and rabies risk.



Rabies post-exposure prophylaxis decision tree poster

(https://www.who.int/southeastasia/news/detail/28-02-2024-rabies-post-exposure-prophylaxis-decision-tree--decide-with-confidence)

PEP decision-making is influenced by various factors including the type of exposure and wound, animal characteristics, as well as the surveillance capacity across sectors. WHO provides comprehensive recommendations for human PEP.

To empower health-care providers on the ground, WHO's recommendations on rabies PEP have been transformed into a visually intuitive decision-tree poster. This poster serves as a quick reference to aid professionals in making informed decisions for timely and effective PEP. It considers various scenarios and can be used along different surveillance capacities. In cases of doubt the options "Unknown" should always be used.

WHO human rabies vaccines global market study – December 2020

(https://www.who.int/publications/m/item/who-human-rabies-vaccines-global-market-study-december-2020)

In addition to providing the most up-to-date understanding of the current and future trajectory of global supply and demand of rabies vaccines for PrEP and PEP, this study identifies actions to help address the anticipated market dynamics and serves as an important resource for the implementation of the Zero by 30 strategy.

Rabies: rationale for investing in the global elimination of dog-mediated human rabies

(https://iris.who.int/handle/10665/185195)

Rabies is a global public health problem with important socioeconomic impacts. Human rabies is preventable; almost all cases are transmitted through the bite of a rabid dog. Elimination of human rabies is possible. Technical support and tools are available. This report covers:

why investment is needed: key rationale;

investment purpose: global elimination of rabies;

investment in action: four case examples in Bangladesh, Philippines, South Africa (Kwa-Zulu Natal) and United Republic of Tanzania; and

summary results of case examples: programme similarities and differences, and health impact success stories from case examples.

Laboratory techniques in rabies, volume 1, 5th edition

(https://iris.who.int/handle/10665/310836)

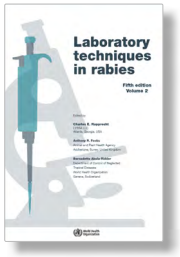
The fifth edition of WHO's laboratory techniques in rabies provides insight into validated methods recommended for use in diagnostic laboratories, but it also includes research. Improved diagnostics will strengthen surveillance of the disease, leading to enhanced control of rabies where it is most needed. Volume 1 includes:

Part 1 – General considerations

Part 2 – Detection of virus

Part 3 – Demonstration of antigens

Part 4 – Demonstration of viral antibodies



Laboratory techniques in rabies, volume 2, 5th edition

(https://iris.who.int/handle/10665/310837)

Volume 2 includes:

Part 5 – Demonstration of viral nucleic acids and sequences

Part 6 – Production of biologicals

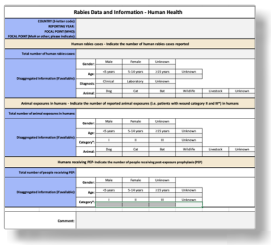
Part 7 – Potency determinations

Sharing rabies data: commit to submit

(https://www.who.int/publications/m/item/sharing-rabies-data--commit-to-submit)

Despite rabies being a notifiable disease in many countries, data collection and surveillance are often neglected.

This flyer explains the 10 reasons for sharing data on rabies and how to submit national data to WHO.



French

WHO rabies data reporting template (Excel)

(https://www.who.int/publications/m/item/who-rabies-data-reporting-template)

Surveillance data – underpinned by effective country reporting and surveillance systems – are critical for evaluating the progress of national rabies control and elimination campaigns and achieving our global goal of zero human deaths by 2030.

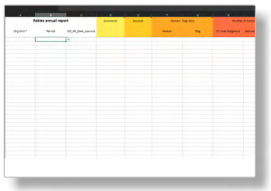
WHO supports and promotes integrated reporting systems for human and animal rabies surveillance.

We invite countries to actively participate in this global effort by nominating a national focal point and sharing their rabies surveillance data, using this template.

DHIS2 rabies data reporting system

(https://www.who.int/activities/improving-data-on-rabies/improving-data-on-rabies)

Rabies surveillance data can directly be shared in WIDP.
To request access to the DHIS2-based data platform including training, submit data you already have collected. For further comments and queries, please contact the zoonotic NTD team at abelab@who.int.



Rabies data reporting template for WIDP upload (Excel)

Rabies surveillance data can further be shared by downloading and completing the rabies data template. This template can either be uploaded into WIDP or shared via email.

Concept Note, News

National bridging workshop on rabies

(https://iris.who.int/handle/10665/376699)

The method used for NBW-rabies adapts exercises and tools from the existing generic NBW workshop sessions and complements the frameworks used for NBWs with rabies-specific key documents and frameworks. The "Stepwise Approach towards Rabies Elimination" (SARE), a much-used monitoring and evaluation framework specifically developed for rabies, constitutes a central element of the NBW-rabies approach.

The method used is very dynamic and interactive, based on group exercises with a gamified approach and user-friendly materials which enable the identification of synergies, the review of gaps and the development of a joint road map between the sectors.

Validate your success – Information posters

We have a tangible opportunity to eliminate the pain, suffering and death inflicted by rabies on humanity.

Use this chance to demonstrate your abilities and strengths by submitting your dossier to get your achievements recognized.

Inspire other countries and celebrate this tremendous milestone on the way to eliminating dog-mediated human rabies. More information is contained in the information posters.

French, Spanish

Template dossier for validation of elimination of dog-mediated human rabies as a public health problem

(https://www.who.int/publications/m/item/template-dossier-for-validation-of-elimination-of-dog-mediated-human-rabies-as-a-public-health-problem)

This editable template dossier was designed to help national rabies programme managers prepare a dossier with supporting evidence for presentation to WHO, requesting validation that rabies has been eliminated as a public health problem.

The information presented in the dossier will help reviewers to understand the achievements of the programme by providing both epidemiological evidence and the broader context.

WHO Academy online courses

French

Rabies & One Health: from basics to cross-sectoral action to stop human rabies deaths

(https://whoacademy.org/coursewares/course-v1:WHOAcademy-Hosted+H0051EN+H0051EN_Q4_2024?source=edX)

Despite being fully preventable, dog-mediated human rabies kills tens of thousands of people every year, especially in rural and impoverished areas in Africa and Asia. This course provides participants with knowledge about the biology and epidemiology of this neglected zoonotic disease, the current "Zero by 30" rabies elimination strategy, and how to prevent rabies in people and dogs by taking a One Health approach. The learning package consists of seven modules, which include downloadable video-lectures by global experts and professionals confronted with rabies in the field.

One Health in action against neglected tropical diseases

org/coursewares/course-v1:WHOAcademy-Hosted+H0020EN+H0020EN_Q3_2024?source=edX)

NTDs continue to cause hardship and harm to more than one billion people worldwide, burdening individuals, families and communities who are already marginalized and disadvantaged. Taking a One Health approach that recognizes the relationships between human, animal and environmental health is key to sustainably addressing these diseases. This course provides practical ideas, tools and examples to enable each of us to take One Health action towards the global goal of substantially reducing the burden of NTDs by 2030.

WHO road map on neglected tropical diseases and companion documents

Arabic, Chinese, French, Russian, Spanish

Ending the neglect to attain the Sustainable Development Goals: a road map for neglected tropical diseases 2021–2030

(https://iris.who.int/handle/10665/338565)

The road map was prepared through an extensive global consultation pursuant to decision EB146(9) of the Executive Board at its 146th session in February 2020 that culminated in the endorsement of the document by the Seventy-third World Health Assembly in November 2020. It sets global targets and milestones to prevent, control, eliminate or eradicate 20 diseases and disease groups as well as cross-cutting targets aligned with the Sustainable Development Goals.

Arabic, Chinese, French, Russian, Spanish

Ending the neglect to attain the Sustainable Development Goals: a framework for monitoring and evaluating progress of the road map for neglected tropical diseases 2021–2030

(https://iris.who.int/handle/10665/341313)

The monitoring and evaluation framework is a companion document to the road map. It provides guidance on mainstreaming the monitoring and evaluation of NTDs within health information systems and emphasizes that these are integral components of interventions against NTDs. The framework presents a set of standardized indicators that will enable comparability and greater accountability among stakeholders, starting at country level.

French, Spanish

Ending the neglect to attain the Sustainable Development Goals: a sustainability framework for action against neglected tropical diseases 2021–2030

(https://iris.who.int/handle/10665/338886)

The sustainability framework companion document is intended to help Member States, stakeholders and partners align their strategies on identifying sustainable pathways towards the road map 2030 targets. This living document will be improved further as experience is gained from the application of these principles in field conditions.

Ending the neglect to attain the sustainable development goals: One Health approach for action against neglected tropical diseases 2021–2030

(https://iris.who.int/handle/10665/351193)

This companion document aims to support a range of stakeholders – including countries in which neglected tropical diseases are endemic, international organizations and non-State actors – to achieve the road map targets through a transdisciplinary, cross-cutting One Health approach. Specifically, it provides guidance on the One Health actions needed by major stakeholders and how to support a paradigm shift towards One Health in national NTD programmes. Examples of common One Health challenges and how they can be overcome as well as illustrative cases studies are provided throughout.

Ending the neglect to attain the Sustainable Development Goals: a global strategy on water, sanitation and hygiene to combat neglected tropical diseases 2021–2030

(https://iris.who.int/handle/10665/340240)

The global WASH strategy companion document supports the achievement of the road map targets. It sets out the critical role of WASH for prevention, care and management of NTDs, and the actions needed to ensure that WASH efforts result in improved and sustained health and well-being outcomes. The strategy builds upon collaborations between both sectors over the past decade and calls for strengthened partnerships to improve access to WASH among populations at highest risk of diseases of poverty.

DHIS2: District Health Information Software (version 2); NBW: national bridging workshop; NTD: neglected tropical disease; PEP: post-exposure prophylaxis; PrEP: pre-exposure prophylaxis; WASH: water, sanitation and hygiene; WHO: World Health Organization; WIDP: WHO Integrated Data Platform.

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